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Monitoring of photovoltaic power plants and comparison of their operational parameters

Diploma thesis

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2. Focus on identification of crucial operational parameters for performance of the PTV plant and utilization of electricity produced.
3. Develop a methodology for monitoring the operation of photovoltaic power plants in order to compare plant performance under different conditions.
4. Perform operational monitoring and evaluation of specific selected power plants.
5. Summarize the operational parameters, compare the data obtained and try to generalize the conclusions.

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1 Introduction

It is quite challenging to satisfy the energy supply these days as the energy demand intensifies, making it a more pressing concern than ever. In addition, increasing environmental concerns of the population and the global drive for optimal sustainable energy sources are not conducive to the continued use of conventional methods of electricity generation. This led to an effort to find new and environmentally sensible approaches to generate electricity a little “greener.” Therefore, the concept of renewable energy sources and their alternatives found its foundation in global energy production.

Production of electricity from renewable energy sources is getting increasingly used as they skyrocket in popularity among various investors and ordinary green enthusiasts. This growing trend is supported not only by green activists but also by the European Union, which aims to be one of the major leaders of the greener future of the world. This trend contributed to the extension and the increase of awareness of sustainable electricity production among the public.

Energy transformation is the key and base principle of obtaining electrical energy. Electrical energy does not just appear, nor is it created from nothing, as someone could suggest, but rather transformed by using specialized devices and facilities. The working principle of such facilities is based on various physical phenomena and laws of physics. These facilities are commonly referred to as power plants, and each can operate on a different power generation base. As already mentioned, the principle of operation of power plants is based on the conversion of energy, which can be, for example, the combustion of several types of fuels such as coal, waste, and fossil fuels, but also by the use of the phenomenon of nuclear fission. On the other hand, there are power plants that are based on generating electricity using renewable sources. These power plants generate energy through a variety of methods, utilizing renewable sources such as biomass, water flow, wind, and, most importantly, solar energy. Each of these sources offers unique advantages in harnessing natural energy. Biomass involves the combustion of organic materials, transforming stored chemical energy into electricity. Hydroelectric power plants convert the kinetic energy of moving water into electrical power, often through dams with installed turbines. Wind energy is “captured” by wind turbines that convert the mechanical force of the wind into electrical energy.

The use of the photovoltaic effect is, nevertheless, one of the most important and progressively developing technologies in energy generation. This process converts sunlight directly into electricity through photovoltaic cells. These cells are made of semiconductor materials and generate an electric current by absorbing photons of sunlight and releasing electrons. The photovoltaic effect plays a critical role in the transition to sustainable energy systems because solar energy is abundant, clean, and becoming more and more affordable.

Many years passed between the discovery of the photovoltaic effect and its practical application in solar panels. We have now found ourselves finally using this energy production on a wider scale. According to Weber (2024), over the past two years, the number of newly connected solar power plants to the power grid has more than doubled in Czechia. As of early 2024, a total of 167,000 solar power plants were connected, including 116,000 new installations over the previous two years. Despite this rapid growth, the energy generated by these solar plants still accounts for only a small percentage of the country's annual energy consumption.

This Diploma thesis aims to deliver some basic information about the working principle of photovoltaic systems and their functionality in different conditions. Reviews the previous implementation of the photovoltaic effect in the past and tries to make the reader familiar with its history and use. The thesis also examines the various parts of solar power plants and introduces them to the reader. The practical part investigates the actual performance of such power plants according to the procedures stated in the methodology part of the thesis. Furthermore, the identification of the crucial performance of each monitored photovoltaic power plant is examined. The last part of this thesis is dedicated to concluding the practical and theoretical parts, where data parameters are measured, obtained, and analyzed according to established methodology.

2 Literary Research

2.1 History of photovoltaics

As Lenardic (2022) stated, the first man who described and discovered the photovoltaic effect was a French scientist named Edmond Becquerel, who found that if certain materials are exposed to sunlight, they can produce a small amount of electric current. This phenomenon was discovered by Becquerel as early as 1839, yet it did not receive further scientific interest for 75 years. In the 1870s, the search continued with the discovery of the photovoltaic effect in selenium by Willoughby Smith. Heinrich Hertz continued the investigation of this phenomenon in solids, specifically in selenium. His research confirmed and revealed the production of electric current under certain circumstances in selenium, resulting in the first selenium photovoltaic cells that were able to convert 1 to 2% of light into electricity. The most extensive and incredibly detailed theoretical analysis of the photovoltaic phenomenon was written by Albert Einstein, who first observed the phenomenon in 1904. He was awarded the Nobel Prize in Physics in 1921 for his theoretical explanation. The next step was taken in 1954 by Bell Laboratories, creating a viable photovoltaic cell on a crystalline silicon basis. These photovoltaic cells had an efficiency of around 4% in converting light into electric current. It was a groundbreaking achievement by scientists, marking the first time solar cells became practical for generating a usable amount of energy.

The major breakthrough came in 1958 with the creation of the first satellite, named Vanguard 1, whose main power source was solar cells. These solar cells managed to operate for around 7 years (Williams, 2022). Between 1950 and 1969, multiple companies started to do their own research in this field. Suddenly, the price of generated watt of energy cost around 1,785 USD, which was not very lucrative. Hoffman Electronics manufactured the first photovoltaic cell with an efficiency of around 9%, which meant a greater breakthrough to the contrast of the first photovoltaic cells. (Lenardic, 2022) Further research was supported by NASA and the US government as “The Space Race” has slowly began (Válek, 2022, pp. 6-10). The use of photovoltaic cells as one of the sources of electrical power found its firm foundation in the field of space exploration because of the invention of radiation-proof silicon solar panels (Lenardic, 2022). Further support from the US emerged in the 1970s in the form of tax acts and the creation of specialized grants as the energy crisis struck the citizens of the states (Chu & Lawrence Taranzo, n.d.). The government was more involved than ever in the support of

photovoltaics in the public sector. The National Renewable Energy Laboratory was founded by the US Department of Energy. According to Smith (2024), this new organization's main task was to supervise and keep an eye on the research on renewable energy sources.

In accordance with Richardson's article (2024), in the 1980s, solar technology commenced further progress with the introduction of new materials like cadmium telluride and amorphous silicon, which led to the improvement of efficiency and reduction of the price of solar panels. Important achievements lured the first investors from both the public and commerce sectors, the same as photovoltaic technology, which became more affordable for enthusiasts from both sectors. In the same decade, the ARCO solar company contributed to the creation of the first 1 MW power plant located in the US state of California. This company also introduced amorphous solar technology in their photovoltaic modules.

A resurgence of interest began between the 1990s and early 2000s. This time period saw a significant revival of attention, sparking fresh projects and advances in the field. Research was mainly focused on enhancing the efficiency of panels, the same as the reduction of their prices. In 1991, efficient photoelectrochemical cells were developed. A year later, a key milestone was reached by scientists from the University of South Florida by constructing a thin-film cell, breaking beyond 15% efficiency for the first time (U.S. Energy Information Administration, 2008). Germany's ambitious plan to build 100,000 solar roof installations started at the end of the decade. One of the primary goals might be to encourage the installation of photovoltaic systems on both residential and commercial buildings with financial support provided for both new and extensive installations up to 1 kW (Smith, 2024). At the end of the decade, in 1999, the world of photovoltaics surpassed a great milestone: "The worldwide production of electrical energy through photovoltaic cells exceeded 1 GWh." (True-South Solar, n.d.). According to Fernández (2024), this outcome was not bad at all; in fact, it was quite good for that period. For reference, the global installed capacity in 2023 was approximately 1,624 GW.

As per the article of Čermáková (2024), the Czech Republic did not stay that behind, as the first photovoltaic power plant was installed at the summit of Mravenečník in 1998. This initial project was intended for educational purposes, with the pivotal aim of supporting domestic Czech manufacturers. Tesla Trimex company based in the town of Rožnov pod Radhoštěm was selected as the main contractor to supply necessary parts for the construction of the

photovoltaic power plant. The company was founded in 1993, and its main specialization was in the processing of silicon for the purpose of use in photovoltaics. The power plant was designed with a maximum capacity of 10 kWp and successfully generated 6,650 kW hours annually. Unfortunately, it was constantly targeted by vandals, who were repeatedly damaging the power plant site. The ČEZ organization, which was responsible for the project, decided to move the power plant to a safer location in 2002 to prevent further damage to the instalment. The project continued at the site of the Dukovany power plant until it was terminated in 2014. Prime Minister Miloš Zeman and his cabinet introduced the “Slunce do škol” project through the State Environmental Fund, offering solar panels for primary and elementary schools at no cost. The project started in 2000 and continued for many years. These instalments’ purposes were mainly educational to introduce and demonstrate the sustainable ways of production of electricity to the students. With an installed capacity of only 100 watts, the plants were not suitable to serve as a direct power source for the schools.

The development of solar energy in the Czech Republic surged in 2009 and 2010 when there was a huge increase in newly built photovoltaic power plants. During this period, the Czech Republic sought to significantly increase the share of solar energy in the overall energy sector and catch up with its Western neighbours. This acceleration was supported by a generous subsidy system set out by Act No. 180/2005 Coll. The legislation introduced guaranteed feed-in tariffs designed to ensure a return on investment within 15 years, with the total period of operational support lasting up to 20 years. Importantly, the law also included a safeguard clause limiting the annual decrease in feed-in tariffs to a maximum of 5%. These conditions created a highly attractive environment for investors, many of whom recovered their initial investment within 7 to 10 years. Relatively large technological advances and the entry of new solar panel manufacturers from China into the domestic market helped to significantly increase the installed capacity from 40 MW to 1,959 MW between 2008 and 2010. This huge growth of interest in solar energy has caused a relative problem to some farmers as new installations started to grow on agricultural land. Fortunately, this is not as common nowadays as it used to be, as the emphasis is on using the roofs of houses, industrial buildings, underused sites, and difficult-to-access locations (Čermáková, 2024). As reported by Solární asociace (n.d.), one of the next photovoltaic booms in the Czech Republic began around 2021 when an energy crisis caused by many factors began. The high demand for new

photovoltaic power plants led to the rise of many new installation companies, driving competition and innovation within the industry. This growth made solar energy more accessible while creating new job opportunities in the renewable energy sector. With a combined peak capacity of 970 MW, 82,799 solar plants were connected to the grid in 2023. This remarkable growth highlights the ongoing pace of solar deployment to boost the production of renewable energy.

2.2 Solar irradiance in photovoltaic

Solar irradiance refers to the power of solar radiation falling on a given area and is measured in watts per square meter (W/m^2). Solar radiation is a whole consisting of all solar energy wavelengths, including light, ultraviolet, and infrared. Solar irradiance directly influences the energy output of solar panels, which is crucial for their performance. Higher irradiance means more solar energy is available for conversion into electricity as more sunlight reaches the photovoltaic cells. In other words, solar irradiance represents the intensity of electromagnetic radiation emitted by the Sun that reaches the Earth's surface, including light in the form of photons. When photons hit the solar cells, they transfer their energy to the material, typically silicon, initiating a process called ionization. This energy excites electrons within the material, freeing them to create a direct current. Geographical location, atmospheric conditions, seasonal variations, and time of day are key factors that influence the intensity of solar irradiance at a specific location. The angle at which sunlight hits the surface also affects the amount of radiation received, especially in the context of solar panels, which are often tilted to optimize this angle (Wenham, Green, Watt, Corkish, & Sproul, 2012, pp. 10-18). Irradiation intensity affects not only energy production but also plant growth, so monitoring it can also be important for farmers.

2.2.1 Photovoltaic effect

This chapter will be delighted with processes connected with the effect of photovoltaics and its working principles, as in the previous chapter, the ionization effect was mentioned. The photovoltaic effect is the outcome process of sunlight hitting the solar cell in a panel. The process takes place within solar cells, which are made up of two distinct semiconductor materials: p-type (positive) and n-type (negative). They are often referred to as an anode (P-type) and a cathode (N-type). The combination of these two materials forms a p-n junction. P-type is a positively charged material with more holes in the valence band, and N-type is its

opposite, with free electrons. The absorbed energy from the photon excites an electron, freeing it from its bond and creating an electron-hole pair. This typically happens when the photon energy exceeds the band gap of the semiconductor. Due to the electric field at the p-n junction, electrons are driven toward the N-type side, and holes toward the P-type side. This separation of charge carriers prevents recombination and allows for current to be generated. The nearby electrons then fill this hole, leaving another hole behind. This movement, under the influence of the internal field, results in a directional flow of charge carriers, thus current flows, and voltage is generated by the potential difference created across the p-n junction. (PennState, 2023). The combination of current and voltage creates an electric power which can be calculated by the equation:

$$P = V * I [W] \quad (1)$$

2.2.2 Types of Irradiances

2.2.2.1 Diffuse horizontal irradiance

As the Sun emits light toward Earth, the sunlight passes through our atmosphere, which acts as an obstacle for the beam of light. The interaction of light with particles and molecules causes a scatter of the beam. This scattering distributes light evenly across the sky, contributing to the creation of diffuse light. The amount of diffuse irradiance depends on factors like cloud cover, atmospheric conditions, and the position of the Sun in the sky (Algora & Rey-Stolle, 2016, pp. 7-12). This scattering can also be influenced by dust, various pollutants, and aerosols, which can be present in the atmosphere (US Department of Energy, 2024). The

DHI



Figure 1: Diffuse horizontal irradiance, source: SOLARANYWHERE (2025)

light that passes through the atmosphere then falls onto the surface from different directions, as shown in Figure 1. As noted by Wenham et al. (2012, p. 7), the sunlight that manages to pass through the obstacles in the atmosphere is reduced by about 30% compared to before it entered the atmosphere.

2.2.2.2 Direct normal irradiance

It is an irradiance that was not scattered by the atmosphere and reached the surface directly. It can be influenced by latitude, season, and weather. It often peaks during midday, when the Sun is at its zenith. The Global Solar Atlas (n.d.) defines zenith as the point at which the Sun is positioned highest in the sky during the day

2.2.2.3 Global horizontal irradiation

Global irradiation is the sum of direct and diffused irradiance. It can be calculated via an equation, where angle alfa stands for the zenith angle (SolarAnywhere, n.d.). According to Chmel and Hamerník (2016, p. 6), diffuse irradiance normally accounts for half of global horizontal irradiance in summer but is substantially greater in winter, accounting for 60%, perhaps due to more gloomy weather in winter.

$$GHI = DHI + DNI * \cos(\alpha) \left[\frac{W}{m^2} \right] \quad (2)$$

GHI – Global Horizontal Irradiance [W/m²], DHI – Diffuse Horizontal Irradiance [W/m²], DNI – Direct Normal Irradiance [W/m²], α – zenith angle [°]

Source: SolarAnywhere. (n.d.). Definitions. Retrieved November 30, 2024, from <https://www.solaranywhere.com/support/data-fields/definitions>

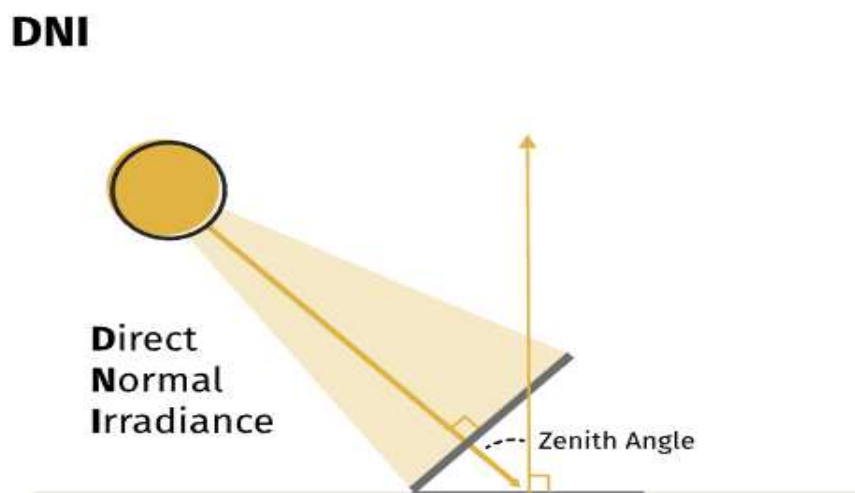


Figure 2: Direct normal irradiance, source: SOLARANYWHERE (2025)

2.2.3 Map of solar irradiation in the Czech Republic

Solar irradiation maps can be important for potential investors who would like to build highly efficient photovoltaic power plants. These maps show the solar energy potential in specific regions and provide a crucial overview of the spatial distribution of solar radiation in the Czech Republic. The level of solar radiation in different areas is influenced by various factors, including geographical location, topography, and climatic conditions. As shown in Figure 3, the most suitable location would be South Moravia, the Zlín region, but also the area around Hradec Králové. If the cloudy days are not counted, the Czech Republic experiences 1,400–1,700 hours of sunlight annually, which corresponds to an average solar irradiance of approximately 950–1,150 kWh/m², depending on the region. Additionally, Figure 3 shows that the Czech Republic receives an average of 949 to 1,168 kWh of solar energy annually, which holds potential for conversion (Isofen Energy, n.d.). This figure will play a key role in the practical part of the thesis, providing a basis for comparing and interpreting the experimental findings.

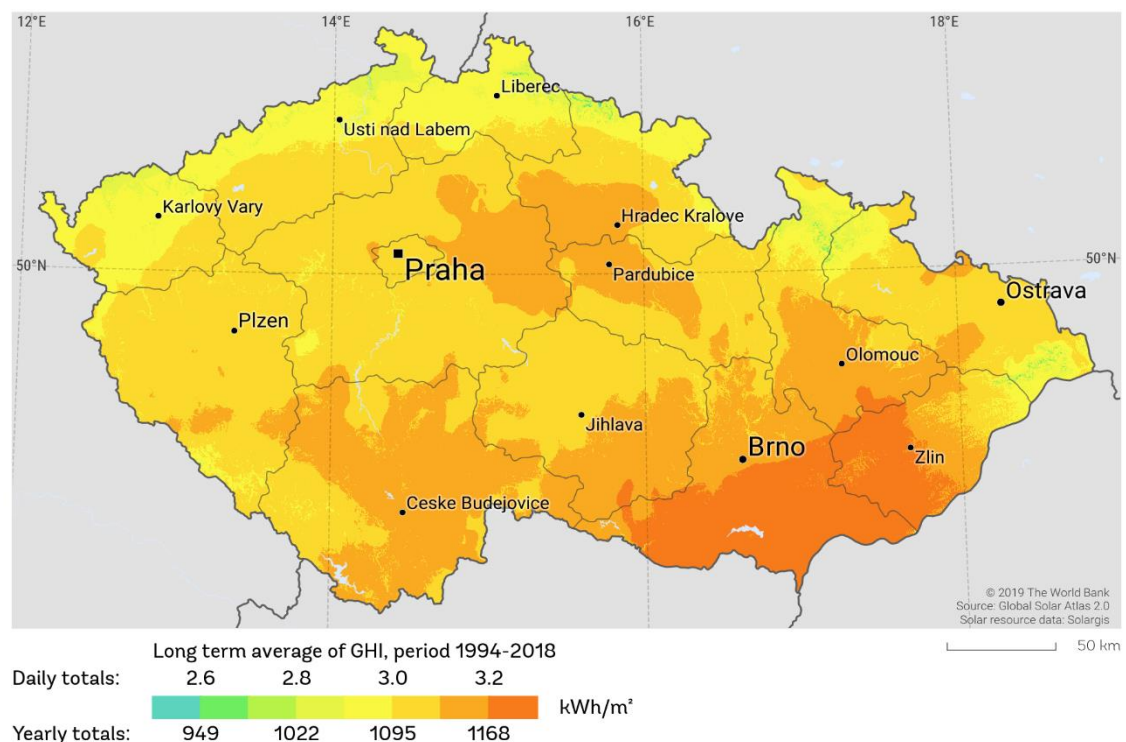


Figure 3: Map of solar irradiation in the Czech Republic, source: Global Solar Atlas (n.d.)

2.3 Parts of a photovoltaic power plant

A solar power plant is a complex system of individual components forming a functional whole. Each component has a crucial role in the process of generating electrical energy. The cornerstone to understanding the overall functionality and performance of an entire system

is to understand the individual components and how they work together. Each component serves its purpose and ensures the smooth operation of the plant and reliable energy supply, from energy collection to energy conversion and further distribution.

This part of the diploma thesis will thoroughly introduce and describe the individual parts and explain their principle of operation and functioning. Provide an insight into the parts such as solar panels, inverters, batteries, and others.

2.3.1 Panels

As explained by Pvknowhow (2024), the photovoltaic panel is the core component of the whole power plant, as its operation principle enables the generation of electrical energy. The panel is composed of multiple interconnected layers of solar cells made from semiconductor materials, mostly silicone. The panel as a whole must be precisely designed to withstand a variety of harsh environments and their external influences, like hailstorms, hurricanes, snow, debris, and dust, which the wind could bring. The aluminium case with attached protective glass secures the protection of the internal part of the panels. This case is in the form of a protective frame that surrounds the whole solar panel. This aluminium material can be enhanced to ensure better protection. One of these enhancements could be the anodization of material to be more scratch-resistant and heat-reflective. The chemical composition of the used material is mainly aluminum with small traces of copper, titanium, iron, and other minerals. It is stated that the frame itself has an approximate lifespan of about 25 years. The encapsulant is a binding layer between layers of the panel, and its primary purpose is the protection of cells from external influences like UV rays and temperature. The most commonly used is Ethylene vinyl acetate in the form of sheets laid in the front and back of solar cells. This material then undergoes the process of lamination, in which it obtains its protective properties. As described by Ecoprogetti (2014), the backside is another protective sheet layer made from plastic. This sheet is electrically insulated and provides an additional layer of protection against water and other adverse influences. The usually plastic junction box, which is attached to the plastic sheet at the back, serves as a connection point between the plant and the panel. It contains connectors and bypass diodes, which mitigate the effects of shading, and the whole box should be designed to be waterproof.

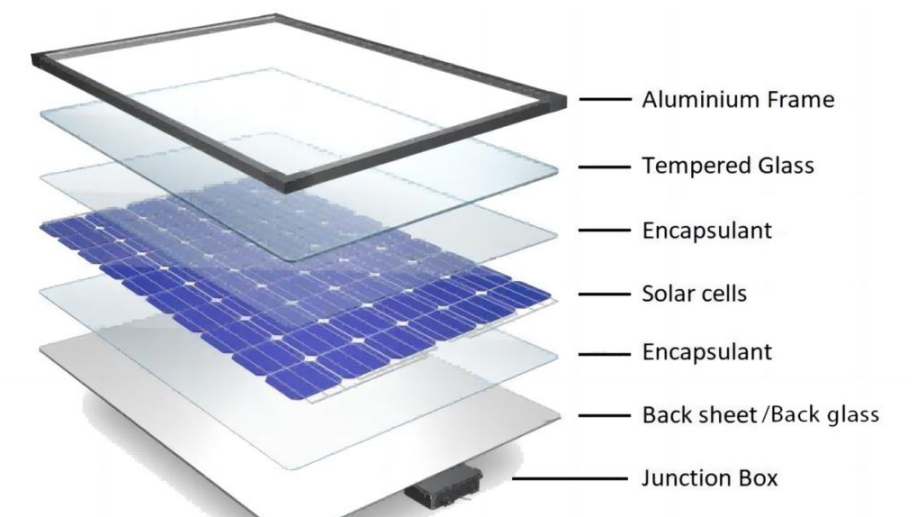


Figure 4: Layers of photovoltaic panel, source: Ibraheem, N. A. K., & Kamoona, G. M. I. (2024)

2.3.1.1 Types of solar cells

2.3.1.2 Monocrystalline cells

As outlined by the Deutsche Gesellschaft für Sonnenenergie (2013, p. 32), monocrystal photovoltaic cells are made from highly pure silicon in the form of ingots, mechanically sliced into wafers with a thickness of around 0.2 mm. The fabrication process ensures that the crystals are cylindrically shaped and then sliced. This is accomplished by the dipping of a seed crystal, which has a defined orientation, into the melted material. The wafers are then chemically cleaned in specialized baths, thus prepared for a diffusion furnace, where phosphorous gas is present. This phosphorous gas creates an n-doped layer on a wafer, thus completing the process as a p-doped layer is created by a layer of boron. The last step is the creation of an anti-reflective coating and the printing out of current collectors. This overall manufacturing process is called the Czochralski process. The structure of thin wafers forms a functional solar cell with an efficiency between 15-18.8 %, with the circuit current between 3 - 16 A. Such cells have a dark blue colour, with possibly a slight overlay in black. Monocrystalline panels are generally more efficient under direct sunlight. However, under cloudy or diffuse light conditions, the performance advantage over other technologies, tends to diminish and may vary depending on the specific panel design and spectral sensitivity.

The production process is rather more expensive than the polycrystal cell because of its relative complexity (Energy Education, 2024). Nevertheless, as emphasized by the American Solar Energy Society (2021), the solar panels with these cells have a long lifetime, around 40 years, and because of the combination of relatively high efficiency, can be

considered a good choice for residential and commercial applications, despite their high purchase price.

2.3.1.3 Polycrystalline cells

As described by the Deutsche Gesellschaft für Sonnenenergie (2013, p. 33), in the manufacturing process, silicon is melted and poured into molds to form a block, which is then cooled down and solidified into the required structure. The boron is added to the melted material to create a p-doped layer. The solidification of blocks helps to create an extensive number of homogenous silicon crystals. After this process, the block is sliced down to the 0.2 mm wafers from which the cells are made. The cleaning process ensures that no dust from slicing remains on the surface. The anti-reflective layer with contacts is added in the same way as for monocrystalline cells. Although this method is relatively cheaper, the performance of such a cell is lower than that of a monocrystal. This is caused by the homogenous silicon crystals, which have a different orientation in the cell. This phenomenon can be observed directly on the surface of the cell, as it causes the cell to develop its specific pattern in blue or grey colour, which can overall resemble a frost-like appearance.

Based on information provided by Janet (2024) and the American Solar Energy Society (2021), the overall efficiency of this type of cell ranges from 13% to 16%. This is primarily due to the high number of crystals, which limit the space available for electron movement. The lifespan of such a panel is estimated at approximately 35 years, but this can differ from manufacturer to manufacturer. They are widely used in installations all around the world, which may just be due to the lower price per panel.

2.3.1.4 Thin film cells

The thin film is another type of solar cell that has been developed since the 1990s. Compared to their predecessors, they are very easy to produce, as the solar cells are made up of three layers. The thickness of these layers is approximately around 0.001 mm, making them highly effective for light conversion. The production process does not require high temperatures, as the deposition temperature is between 200 and 600 °C (Deutsche Gesellschaft für Sonnenenergie, 2013, p. 49). The Photovoltaic material is laid on the base, with the conductive layer and protective layer, which is usually made of glass or plastic. Solar panels manufactured in this way are very flexible, lightweight, and relatively cheap. As Peake (2018, pp. 118-140) explains, the three main types of thin-film solar cells include amorphous

silicon, cadmium telluride, and copper indium gallium selenide, each with distinct characteristics and applications.

As outlined by the American Solar Energy Society (2021), the Amorphous silicon cells are ideal for cost-effective applications as they are quite easy to manufacture. The Cadmium Telluride has a better conversion ratio, but the cells are not widely used as the Telluride is exceedingly rare and also because of the high toxicity of cadmium. The last option is Copper Indium Gallium Selenide cells, which are made of several layers consisting of copper, gallium, selenide, and copper. Thin-film cells are considered moderately efficient, but it depends on the specific application and the associated conditions in which they are used.

Based on information from Deutsche Gesellschaft für Sonnenenergie (2013, pp. 49-50), thin film solar cells typically achieve relatively low efficiencies, with maximum levels reaching only around 12%. However, they offer distinct advantages in certain conditions. For example, these cells excel at capturing diffuse light, making them particularly well-suited for areas with low sunlight or cloudy weather. A major advantage of thin-film solar cells is their ability to maintain stable performance even at high temperatures. Unlike other types of solar cells, whose efficiency tends to decrease, thin-film solar cells hardly change their efficiency with increasing temperature. This characteristic makes them ideal for use in hot climates or environments where temperature control is challenging.

2.3.1.5 Generations of solar panels

There are three distinct generations of solar panel development, each of which has seen advances in technology, cost-effectiveness, efficiency, and materials. These generations are defined by the types of materials and manufacturing processes used. Each generation represents a significant step forward in the development of photovoltaics and contributes to improvements in manufacturing processes, energy conversion efficiency, and overall system performance.

As noted by Anissimov (2024), the initial generation of solar cells is known for its high efficiency but also for its significant cost. These cells are produced using highly purified silicon and utilize a single junction to transform photons into electrical energy. In 2007, these first-generation solar cells dominated the worldwide market, making up 89.6% of it. However, their prevalence has decreased over the years. The production processes for these cells are

inherently costly, leading to a prolonged payback period for the initial investment. Due to these high manufacturing expenses, it is unlikely that first-generation solar cells will become cost-competitive with energy sources based on fossil fuels. Despite their high costs, first-generation solar cells have played a crucial role in the development of solar technology. They have set a benchmark for efficiency and reliability, which has driven further research and innovation in the field.

As further described by Anissimov (2024), second-generation solar cells, developed during the 1990s and early 2000s, are defined by their affordability and lower efficiency compared to first-generation cells. These cells predominantly utilize thin-film technology, which relies on minimal material usage and cost-efficient manufacturing methods. Common materials used in their production include copper indium gallium selenide, cadmium telluride, and amorphous silicon. A prominent example of this technology is the solar cells produced by companies such as Nanosolar, which leverage high-speed printing techniques for efficient production. Despite their relatively low conversion efficiency, typically between 10% and 15%, the significantly lower production costs compensate for this drawback. This affordability positions second-generation solar cells as potentially more cost-effective, making them an attractive option for widespread use. Beyond their economic advantages, thin-film solar cells offer remarkable flexibility due to their design. They can be seamlessly integrated into various surfaces, such as building materials or portable devices, expanding their applications beyond conventional solar panels. With continuous advancements in materials and manufacturing processes, these cells are steadily improving in performance and becoming even more cost-efficient over time.

Third-generation solar cells, according to Anissimov (2024), is still largely in the research and development phase and has not yet been widely tested or commercialized. This generation is primarily theoretical and has garnered significant interest from scientists due to its potential to revolutionize solar technology. The main goal of this generation is to reduce the price while increasing the efficiency of the panels. The primary ones are nanostructured and multifunctional cells, which are likely to be used sometime in the near future. The theoretical efficiency could range between 30% and 60% according to Anissimov. The third generation is rather theoretical and currently attracts the interest of scientists as it has not been tested on a wide scale.

2.3.1.6 Lifespan and maintenance of solar panels

As reported by Huawei Technologies (2024), solar panels typically have a lifespan of 25 to 30 years, with some manufacturers providing warranties for an additional 20 to 25 years. A number of variables affect the lifetime, such as the technology used, the surrounding environment, and the quality of the installation of the panel. Most solar panels lose moderately between 0.5% and 1% of their efficiency per year as their performance deteriorates over time. A large number of installed solar panels operate at 80-85% of their original efficiency even after 25 years. Future technological developments, particularly in the materials used in solar cells, could further improve the lifetime and durability of panels.

The lifespan of solar panels is influenced by several key factors, including the quality of the materials used, the installation process, environmental conditions, and regular maintenance. High-quality panels should be subject to strict quality control standards and be designed to withstand adverse weather, various external influences, and UV radiation, as it was previously stated in the thesis. Conversely, lower quality panels, although initially cheaper, can degrade more quickly, leading to reduced efficiency and more frequent maintenance over time, or even a faster end of life for the entire installation. Another issue that can affect the life of the panels is initial poor installation. This can include improper and unprofessional installation, insufficient ventilation, and therefore insufficient cooling of the panels. All of these factors can cause damage or lead to long-term degradation over time. In addition to these problems, issues such as micro-cracks or water ingress can occur due to improper installation or poor manufacturing by the manufacturer. Another problem can be caused by extreme weather, which can be unpredictable and uncontrollable. Possible damage could also be caused by high winds, hail, or rapid temperature fluctuations. In addition, areas of high humidity or exposure to salty air can also be damaging and can cause corrosion in the long term, especially if the protective coatings on the panels are damaged or scratched.

As emphasized by Lam and Gobler (2024), proper maintenance of solar panels is very important to ensure their longevity and optimal performance. The panels themselves do not need any major maintenance, there are just a few aspects that need to be taken care of when necessary. The most important task is cleaning the panels, as dust and dirt can reduce their efficiency. Cleaning is especially important in the winter when snow build-up needs to be removed, or in the dusty summer season as dirt can easily accumulate on the panels. It is

recommended to clean the panels at least twice a year to prevent excessive accumulation of dust and dirt. In addition to cleaning, it is also important to carry out regular inspections of the system. Although solar panels require minimal maintenance, as already mentioned, at least an annual inspection is recommended, for example, checking the grounding of the panels, visually inspecting the connecting parts, or checking for any defects on the panels themselves, etc.

2.3.1.7 Connection of solar panels in the system

There are two main ways to connect solar panels. The first is a series connection, in which the open-circuit voltages of individual panels add up, resulting in a total voltage higher than that of any single panel. Meanwhile, the current flowing through the entire series of panels remains constant when in steady-state conditions. This means that while the voltage increases, the overall current output does not vary, making series connections useful for applications requiring higher voltages. Additionally, if a single panel in a series fails or is disconnected, the entire circuit is interrupted, compromising the integrity of the whole system. This highlights the critical importance of the functionality of each panel (Energy Nest, 2020, pp. 72-76). One advantage of series wiring is that fewer wires are needed. It is also more effective for transmitting power over long distances with minimal loss. However, as it was mentioned, not only malfunction but also shading of one panel can reduce the performance of the entire string, as the weakest panel affects the current of the whole system. Consequently, if one panel is less efficient than the others, the overall system efficiency decreases (Brentley, 2024). Voltage and power output in a series connection can be calculated by use of formula 3 and 4, which are below:

$$V_{total} = V_{panel} * \text{number of panels in connection} [V] \quad (3)$$

$$P_{total} = V_{total} * I_{panel} [W] \quad (4)$$

On the other hand, an alternative connection is a parallel connection. In this way of connection, the voltage across all of the panels stays constant while the current generated by each panel builds. This implies that as more panels are added in parallel, the overall current output increases. Furthermore, if one panel fails in a parallel connection, the other panel can continue to operate, assuring the circuit's effectiveness as each panel operates independently (Energy Nest, 2020, pp. 42-45). Also, shading on one panel does not affect the performance

of the others. This makes the system more efficient, especially in areas with inconsistent sunlight. However, setting up parallel connections is more complex, requiring additional components such as junction boxes, which can increase installation costs. Wires with larger cross-sections are necessary as higher currents flow through this system (Brentley, 2024). Current and power output in a series connection can be calculated by use of this formula:

$$I_{total} = I_{panel} * \text{number of panels in connection} [A] \quad (5)$$

$$P_{total} = V_{total} * I_{panel} [W] \quad (6)$$

As explained by Brentley (2024), the last option is a hybrid wiring configuration, which blends series and parallel connections to optimize both voltage and current. This method is commonly used in larger solar systems to maintain a balanced ratio of voltage and current.

Regardless of the wiring configuration, the voltage of the solar panel array must match the inverter's input range. If the voltage is too low, the inverter will not function properly. If the voltage is too high, it can damage the inverter or shorten its lifespan. Additionally, the strings should have roughly the same output voltage. The calculation of a hybrid system is relatively simple, as the panels are connected in series in each string, and the entire power plant consists of multiple strings connected in parallel. It can be calculated by these calculations:

Series connection for each string in the power plant:

$$V_{string\ 1} = V_{panel} * \text{number of pannels in string 1} [V] \quad (7)$$

$$I_{string\ 1} = I_{panel} [A] \quad (8)$$

Parallel connections of each string in the power plant:

$$V_{total} = V_{strings} [V] \quad (9)$$

$$I_{total} = I_{string\ 1} + I_{string\ 2} + I_{string\ 3} [A] \quad (10)$$

$$P_{total} = V_{total} * I_{total} [W] \quad (11)$$

2.3.1.8 Recycling of panels

Many installed photovoltaic panels may already be at the end of their useful life, as they may have been installed decades ago, and their efficiency has significantly decreased over time. As a result, their current performance may no longer be economically viable, especially when compared to modern solar panels with higher efficiency and conversion ratio, prompting the need for replacement, repurposing, or recycling strategies to address this growing issue. Solar panels are made up of several high-cost materials like silver and high-quality silicon. According to Crownhart (2021), it is estimated that by 2050, when panel rectifications are expected to peak, these recyclable resources could be worth approximately \$2 billion yearly. The total volume of waste is expected to reach up to 80 million tons in the same year, which is linked to the fact that not all the material contained can be reused. The French company ROSI Solar is trying to make progress in recycling these panels. Its innovative solution for recovering the valuable materials mentioned above is to use a chemical recycling method that should enable more efficient recovery of these materials. Unlike traditional mechanical recycling, this method should be much cleaner and more environmentally friendly and allow for better recyclability of other parts of the panels. However, this method of recycling may be uncertain as the economic situation intervenes, and it is not certain that this method would be financially sustainable given the falling prices of input materials.

As outlined by Solar Panels Recycling (2024), conventional recycling methods could lead to unwanted pollution in the form of toxicity, as these processes could leak cadmium or lead into the soil and subsequently into water sources or pollute the air. This, of course, depends on whether landfilling or incineration methods are used, both of which are extreme and environmentally damaging alternatives. Recycling helps to mitigate the environmental impact and using recycled materials from previous panels in the manufacture of new panels can reduce the carbon footprint. The aluminium frame and protective glass are some of the simplest components to recycle, as they can be efficiently separated and reused through standard mechanical processes. In contrast, solar cells require high-temperature processing, where they must be melted to extract silicon. This process often involves the use of advanced furnaces that can precisely control temperature to recover silicon with minimal degradation. Meanwhile, other components, such as the polymer encapsulant, require chemical dissolution using solvents that break down the polymer bonds, allowing for the separation.

2.3.2 Optimizers

According to SOLARMAN (2024), optimizers are designed and constructed to maximize the performance of the entire power plant by controlling individual panels. These devices attempt to compensate for variable problems such as shading, dirt build-up, or differences in panel performance. Through their design features, they simply try to maintain the maximum efficiency of the whole system. This chapter will focus on hardware optimizers that are physically connected to the panel rather than software ones that tend to be implemented in an inverter.

At the module level, these optimizers carry out maximum power point tracking (MPPT), guaranteeing that every panel runs as efficiently as possible. The main benefits of module-level optimizers include their ability to reduce mismatch losses, improve energy yield, and enable real-time monitoring of module-level performance. This is particularly important in installations where partial shading, uneven module orientation, or different module types are present. The main advantage of the optimizers is their ability to automatically shut down or disconnect the current flow in the event of a system or component failure, ensuring that no energized direct current remains in the conductors. This capability increases the general safety of the plant and reduces potential fire hazards. This feature, often referred to as Rapid Shutdown Capability, is in compliance with safety requirements in many countries and is often a necessity for larger installations. Either way, this hardware definitely provides greater peace of mind for system owners and installers. Each optimizer is mostly equipped with four separate connectors. Two of these connectors are used to link the optimizer to one or two solar panels, while the remaining two connectors are for connecting the optimizer to the string, enabling it to integrate with other optimizers in the system. This configuration allows for proper communication and synchronization of the entire photovoltaic system..

There are many companies producing optimizers, but two notable examples that are frequently installed are SolarEdge and Tigo. The SolarEdge power optimizer connects to each solar panel individually and tries to ensure optimal power generation. In addition to maximizing efficiency, it also provides in-depth power monitoring capabilities for each individual module. The optimizers are waterproof and meet all other prerequisites to be used even in harsh climatic conditions. Tigo Energy offers a range of optimizers as part of its broad TS4 platform offering. All the optimizers offered are versatile and customizable, offering

different options depending on the specific needs of each power plant project. They are designed for power output between 500 W and 700 W. Tigo optimizers, in particular the TS4-O UHD module, are designed to solve problems such as shading and mismatch of individual panels. The TS4-O model is designed for the best optimization to ensure the highest performance of the photovoltaic panel. TS4-S stands for safety as it is optimized for separate rapid shutdowns of individual solar panels in the system in case of emergency. TS4-M for monitoring and provides a modular approach that allows the user to pick and choose the features needed for each system. The manufacturer guarantees the lifespan of the product for up to 25 years.

SolarEdge Power Optimizers are known for their high efficiency and seamless integration with SolarEdge inverters. On the other hand, Tigo Performance Optimizers stand out for their flexibility and customizable solutions that can be tailored to different system needs. Together, these companies offer some of the most reliable and widely used optimizers in the industry, making them a favourite among installers and system owners who want to get the most out of their solar installations (Tigo, n.d.; Krannich, 2020). In Figures 5 and 6, the optimizers of each company can be seen. Figure 6 shows an optimizer that can control up to two different panels.



Figure 5: SolarEdge optimizer, source: SolarEdge (2024)



Figure 6: Tigo, two-panel optimizer, source: Tigo Energy (2024)

2.3.3 Inverters

As explained by Solar Magazine (2021), the main task of an inverter is to convert direct current electricity generated by solar panels into alternating current electricity. In addition to their main function of conversion, inverters can also perform a part of their function as power optimizers and influence the lifetime of the system by their control of the whole system. The inverters, as well as the optimizers, are equipped with MPPT technology, which, as already mentioned, allows the system to operate at maximum efficiency. Unlike an optimizer, which only controls a specific solar panel, an inverter controls entire strings of panels and can switch them off individually. This feature is particularly important when there are fluctuations in sunlight, for example, due to the passage of clouds or time of day, in order to optimize energy output. Inverters provide comprehensive system performance monitoring. In the case of microinverters and systems with installed optimizers, they are able to track the power conversion rate, power consumption, system status, and even specific issues with individual panels. Solar inverters are also equipped with various safety mechanisms, such as arc fault detection and ground fault protection. These various features should prevent damage to the system and reduce the risk of fire. Most of solar inverters are grid-tied, which means that they not only convert solar energy for indoor use and charge the connected batteries in the system but also supply excess energy back to the grid. The Inverters must comply with grid standards to ensure that the energy sent back matches the grid's frequency and other essential parameters for compatibility. It is vital for an inverter to maintain stable voltage, frequency, and other parameters to ensure proper synchronization with the external grid when the system is connected. Some inverters include built-in features that allow the use of net metering programs.

As outlined by the Deutsche Gesellschaft für Sonnenenergie (2013, p. 121), there are several types of inverters commonly used in photovoltaic systems, including microinverters, central inverters, and hybrid inverters. These devices can also be categorized as either grid-connected or stand-alone, depending on the type of system they are integrated into. Smaller installations up to 5 kWp are usually built as single-phase systems, for example, for water heating or boilers, while larger installations typically use a three-phase system. A single-phase inverter can also be used for three-phase systems; however, this requires two additional single-phase inverters to complete the conversion into alternating current. This setup ensures

compatibility with three-phase systems but may result in higher costs and slightly reduced efficiency compared to using a dedicated three-phase inverter. The three-phase system is a rather better option in terms of efficiency as converted energy is equally disturbed all over three phases.

As explained by the Deutsche Gesellschaft für Sonnenenergie (2013, pp. 133-135), inverters facilitate grid monitoring and provide functions such as displaying operating data, signalling, and data collection. The data can be accessed through a display or a computer, provided the inverter is configured for data transmission, such as via Ethernet, CAN-Bus, USB, or other connection technologies. The computer can analyze the collected data by communicating with the data logger, which enables centralized evaluation across multiple inverters within the same system. This setup not only streamlines monitoring but also provides real-time insights into system performance. Additionally, it allows for automated alerts and notifications to be sent directly to your phone or other selected devices, ensuring that you stay informed about any potential issues or system malfunctions, allowing for faster response times and better overall management.

As further explained by Wenham et al. (2012, pp. 104-105), inverters can also be categorized into light, medium, and heavy-duty types based on their capacity and intended applications. Light-duty inverters are perfect for ordinary house installations because they are made for smaller residential systems with power demands of up to 10 kW. Medium-duty inverters can manage power loads of up to 20 kW, making them appropriate for bigger residences or small business settings. On the other hand, heavy-duty inverters are ideal for industrial applications or large-scale solar power plants since they are designed to withstand high power demands, usually handling loads of 60 kW or more. These inverters typically have an efficiency ranging from 80% to 85%, operating effectively across loads ranging from 25% to 100% of their rated capacity. Efficiency may vary depending on the load, with peak performance generally achieved at higher load levels.

2.3.3.1 Microinverters

Microinverters are small devices mounted on each individual solar panel, converting direct current to alternating current at the panel level. This setup removes the need for a central inverter, enabling each panel to function independently. The total price can reach a higher value as an inverter has to be installed at each panel. The layout of a microinverter is shown on the right side of Figure 7.

As noted by Thoubboron and McDevitt (2024), microinverters offer several advantages, particularly in terms of panel-level monitoring, which enables tracking of the performance of each solar panel individually, making it easier to spot any potential issues. Additionally, microinverters provide greater flexibility in how panels are arranged, making them ideal for roofs with varying orientations, slopes, or shading from nearby trees or structures. However, microinverters come with a few drawbacks. They are generally more expensive than Central string inverters and can be harder to maintain due to their placement on the roof. They also tend to have a lower power output capacity than the panels, which may result in "clipping" — a situation where the inverter cannot manage the full energy output of the panel. Despite these challenges, microinverters are an excellent choice for homes with complex roof layouts. They offer easier system expansion, enhanced performance under challenging conditions, and longer lifespans, with many models backed by warranties of up to 25 years.

2.3.3.2 Solar String Inverters

As explained by Beite (n.d.), string inverters are primarily used in small-scale installations. Individual solar panels are connected in a string, and each panel's DC power is fed into a solar string inverter, which transforms the array's total DC electricity into AC electricity. Solar inverters typically handle the DC output of several solar panels, often 10-15 panels, which are connected in a given string. Conversion rates for contemporary string inverters can reach up to 98%, demonstrating their high efficiency. The performance of all connected panels may be lowered if one of the panels in the string is shaded, though, because they are sensitive to shading. The whole string of panels will also stop generating electricity if the inverter fails. The ideal places for these inverters to be installed are those with steady sunlight and little shade. It is a very simple type of inverter, usually the least expensive option, so they are being widely installed in residential installations. The layout of the solar string inverter can be seen on the left side of Figure 7.

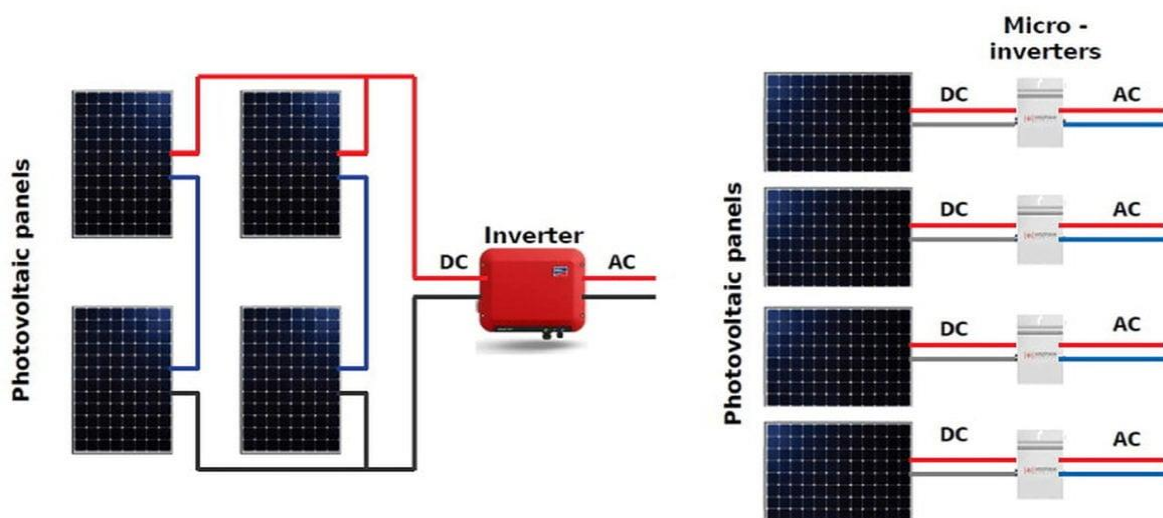


Figure 7: Solar string and Micro inverter layout, source: Gupta U. (2021)

2.3.3.3 Central inverters

As noted by Novergy (2024), central inverters convert a direct current generated by multiple strings of solar panels into an alternating current. They are particularly well-suited for large-scale solar installations, such as solar farms or commercial projects, where they manage high-power conversion capacities. Some models of central inverters can manage up to 500 kW of power, processing energy from as many as 16 strings of interconnected solar panels. These strings are then connected into the combiner box, which serves as a central hub where the output from multiple solar panel strings is gathered. Central inverters require

substantial space for installation, due to their size, and are often located in dedicated rooms or outdoor cabinets designed to withstand harsh environments. The centralized nature of these inverters means that they control power for the entire system. However, this configuration also creates a critical vulnerability: if the central inverter fails, it can lead to a complete shutdown of the system. In such cases, the entire solar plant ceases to generate electricity, as all the panels are reliant on that single unit for DC to AC conversion. One of the primary advantages of central inverters is their cost-effectiveness, particularly for large solar power plants. Their cost per watt is typically lower compared to smaller inverters like microinverters or string inverters, making them an economical choice for large-scale installations.

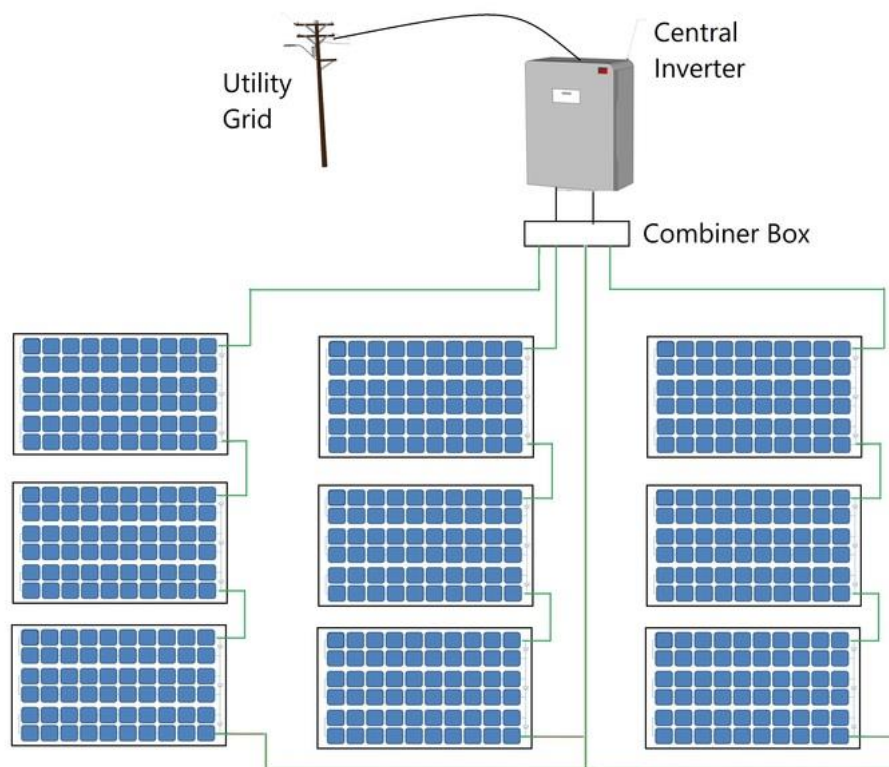


Figure 8: Central inverter layout, source: EcoFlow (2024)

2.3.3.4 Master-Slave Inverters

As described by the Deutsche Gesellschaft für Sonnenenergie (2013, p. 138) and Namkoo Solar (2023), the master-slave inverter configuration is a concept similar to central inverters but with the added benefit of modularity and scalability. In this system, there is a central master inverter that coordinates the overall operation, while slave inverters (smaller inverters) work alongside the master inverter to process the power generated by solar panels.

The master inverter controls and monitors the slave inverters, allowing them to work together as a unified system. The system can be expanded by adding more slave inverters as needed, depending on the size and requirements of the solar installation. This modular design allows for increased flexibility and easier management of large-scale solar systems. The master inverter typically manages the overall power conversion process, in meaning that the slave inverters handle power conversion for smaller sections of the system. In addition, the master inverter typically operates autonomously when the power demand is low, handling the power conversion and distribution by itself. However, as the energy demand increases and exceeds a predefined threshold, the master inverter activates the slave inverters to assist in power conversion.

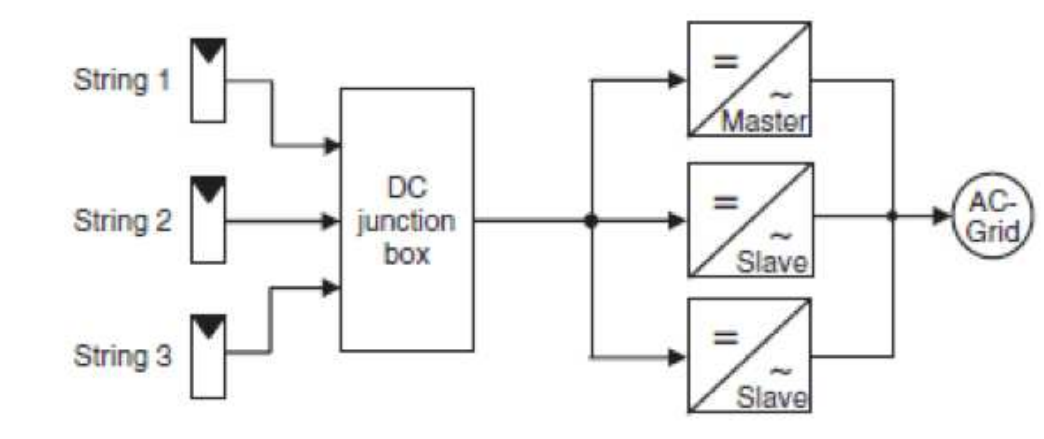


Figure 9: Master-Slave inverter layout, source: Penzo S. (2017)

2.3.3.5 Hybrid inverters

As described by Thoubboron (2019), hybrid inverters are designed for solar systems with battery storage, serving as both an inverter and a battery charger. By combining these two functions into one device, hybrid inverters simplify the system and eliminate the need for a separate battery charger. A key advantage of hybrid inverters is their ability to integrate energy storage, offering flexibility for future upgrades if the system does not have an initially installed battery. In addition, hybrid inverters provide resilience during power outages. When paired with a battery, they can continue supplying electricity to the home even if the grid is down, as some inverters do not have to rely on the grid. Many hybrid inverters also include centralized monitoring systems, allowing users to track both solar panel and battery performance through a single platform.

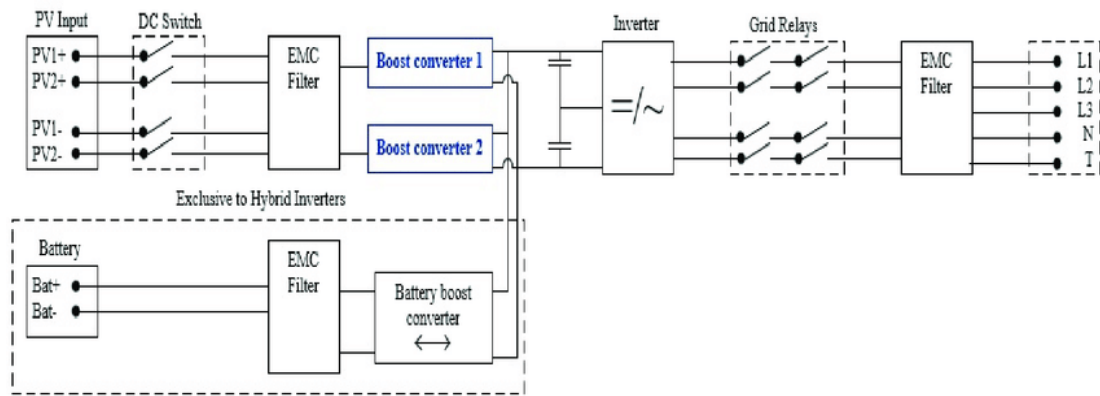


Figure 10: Hybrid inverter, layout of the system, source: Remoaldo, D., & Jesus, I. (2021)

2.3.4 Batteries

The main purpose of batteries in a solar array power plant is to store the generated electrical energy in some usable form. The energy that cannot be used directly in the facility or home where the plant is located can be sold to the grid or stored in a battery for later use. This also depends on the topology of an installed system; for example, if the system is connected as an islanded connection, it is not possible to sell surplus to the network as the system is not connected to any external grid network. However, this is taken care of by the specified system controller or inverter, which serves as a battery charger, but also determines when the excess energy produced will be sent to the external grid. As the energy is produced during the day, when the household does not may have such a high energy demand, the use of a battery is in place. The system starts to charge the battery so the energy produced during the day can be used in the evening. Common battery types available for storing electrical energy include lead-acid, lithium-ion, and AGM batteries, as noted by Fuelcell Store (2024), which outlines the most widely used technologies in the current market.

Different connections of battery cells allow us to either increase the capacity of the system or increase its voltage. That is because of the parallel or series connection of each terminal of the battery. When batteries are connected in a series, the positive terminal of one battery is linked to the next battery's negative terminal. This connection increases the overall system voltage while the capacity in kWh remains the same. Therefore, we may create a 24V system by connecting two 12V batteries. In a parallel connection, all the positive poles of the batteries are connected together, the same as the negative ones. In this way, the overall capacity of the system is increased while the voltage remains the same.

2.3.4.1 Lead-acid battery

The most commonly used and oldest battery is based on lead-acid, with their common power range between 0.1 kWh and 100 kWh. They have high efficiency and can be charged even with a small current. The nominal voltage of one cell is around 2 V, and they are connected in series inside the battery to obtain a 12 V block. Between the cells is diluted sulphuric acid, which facilitates the flow of ions between submerged positive and negative plates of the battery. One of the main advantages of lead-acid batteries is their lower cost compared to other battery types, such as lithium-ion. They are also known for their reliability and ability to handle many charge and discharge cycles. Much of this information is summarized from Deutsche Gesellschaft für Sonnenenergie (2013, pp. 150-151), which explains that these batteries operate using an electrochemical reaction that converts lead sulfate (PbSO_4) to lead (Pb) and sulfuric acid (H_2SO_4) during charging. During discharge, lead and sulphuric acid are again converted to lead sulphate and water, releasing the stored energy.

According to Bowden and Honsberg (2024), higher energy losses may result from their reduced charging and discharging efficiency. Lead-acid batteries have a limited lifespan as well; depending on use and care, they usually last three to five years. Sulfation and loss of active material, which is influenced by the charging and discharging modes, the depth of discharge, and the average temperature during the battery's lifetime, are one of the main causes of battery capacity deterioration.

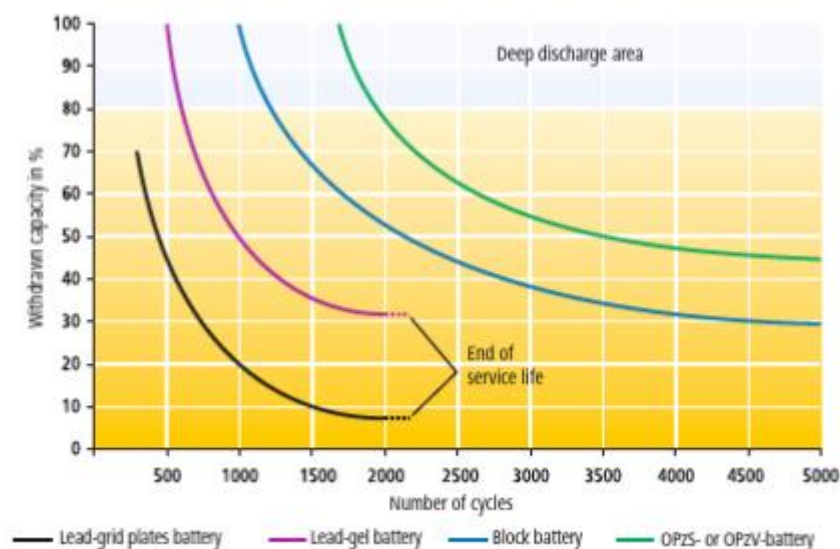


Figure 11: The lifespan of batteries, source: Deutsche Gesellschaft für Sonnenenergie (2013)

The battery life is estimated in a number of charging and discharging cycles. In other words, battery life refers to how many cycles of charging and discharging a battery it can withstand before its capacity drops to a certain point. This point is usually considered to be about 80% of a battery's initial capacity. The battery that reaches this point will still be functional, but the risk of failure in the form of a short circuit will rise higher. The number of cycles varies depending on the type of battery. This issue is shown in Figure 11.

2.3.4.2 Lithium-ion battery

As noted by Wigness (2023), this type of battery has become the most used type of battery in just the last few years. One of the primary explanations for why lithium-ion batteries are a viable option for energy storage applications is their higher charge and discharge efficiency, which is one of their many notable advantages. This efficiency is somewhere between 90–95%. Other benefits are high energy density, long service life, and minimal maintenance. The lifetime is more than 10 years, which can be considered a big advantage over lead-acid batteries, and at the same time, the higher price can be considered more acceptable. Despite being a newer technology that relies on materials in great demand, they are frequently quite expensive. The wrong installation of the battery can cause a severe problem, as some types can be prone to overheating, which could lead, in some cases, to thermal runaway. In other words, the battery could easily catch fire and cause unwanted problems.

Lithium-based batteries differ as there are many subtypes of batteries, like lithium iron phosphate, lithium manganese oxide, lithium nickel manganese cobalt oxide, or lithium titanate-based. Nickel and cobalt lithium base batteries are harshly discussed in Western countries as they are commonly mined by non-ethical and non-ecological methods.

Lithium-iron-phosphate (LiFePO_4) batteries have a long lifespan as they can endure 2,000 to 10,000 charging cycles and have internal chemical stability. This information, along with the following characteristics, is detailed by McQuilter (2024). Chemical stability ensures a lower risk of overheating and fire as they do not contain hazardous materials such as cobalt, making them more environmentally friendly. They have high charge and discharge efficiencies, often over 95%, and allow deep discharge up to 80%. On the other hand, their energy density is lower than some other lithium-ion batteries, such as lithium nickel manganese cobalt batteries.



Figure 12: Pylontech LiFePO4 battery, Pylontech (n.d.)

Lithium-nickel-manganese-cobalt oxide (LiNiMnCoO_2) batteries offer a balance of energy density, lifespan, and safety. This overview, including the specific roles of nickel, manganese, and cobalt, is presented by Taylor (2022). The nickel increases the energy density, thus improving storage capacity, but on the other hand, it has low stability. Manganese improves the thermal stability and safety of the battery. The cobalt inside the battery enhances energy transfer capacity. The main problem with these batteries is cobalt because, as was previously mentioned, it can be mined in unethical ways and is also quite expensive.

Lithium-manganese oxide batteries are known for high thermal stability, thus making it a safer option compared to other types of batteries, as there is a reduced chance of thermal runaway. UFINE (2024) highlights that this battery type is also cost-effective due to the abundance and low cost of manganese. However, lithium mangan oxide batteries typically have a moderate energy density, which is lower than for Lithium Nickel Manganese Cobalt Oxide batteries.

2.3.4.3 Sodium-Nickel Battery

As noted by the European Association for Storage of Energy (2016), these batteries, sometimes called ZEBRA, are typically used in applications where stable and durable energy storage is required, including off-grid systems and backup power. Their high energy density and long-term reliability make them suitable for large-scale energy storage solutions. One of the main properties of sodium-carbon batteries is that they operate at higher temperatures, which are up to 350 °C. Maintaining such a high temperature is important in order to keep the sodium electrodes in a liquid state at all times and prevent them from solidifying. The overall heating and temperature maintenance is provided by an integrated system in the battery itself. The efficiency of this battery is 85-95%, with a life duration of about 15 years. The thermal stability, together with the relatively good lifespan, which is often claimed to be around 4500 charge cycles, makes this type of battery well utilized and suitable for extreme thermal conditions. In addition, sodium is a more accessible and cost-effective base material than lithium, making it a good alternative to lithium-ion batteries.



Figure 13: Sodium-nickel-chloride Battery, source: Marco P. (2010)

2.3.5 Charge controllers

As discussed in Deutsche Gesellschaft für Sonnenenergie (2013, pp. 161-165), the core task of charge controllers is to regulate the flow of energy from the solar panels to the battery storage, in contrast to safety and efficient charging. Without a charge controller, the battery could be overcharged or deeply undercharged, which can cause damage or significantly shorten its lifespan. To prevent such issues, charge controllers are equipped with additional integrated safety features, such as short-circuit protection, reverse polarity protection, and

deep discharge protection. These controllers continuously monitor the state of charge (SOC) of the battery to optimize the charging process and enhance its longevity. To manage disconnection in case of faults or risk of deep discharge, charge controllers often use actuators composed of MOSFETs and IGBTs. Additionally, to address battery ageing, some charge controllers include a pair of microcontrollers that help manage the proper charging and discharging levels, further extending the battery's lifespan. Charge controllers can automatically manage battery maintenance, including using float charging to counteract the self-discharge of a fully charged battery. Some models also feature Maximum Power Point Tracking (MPPT), which optimizes energy efficiency by adjusting charging parameters, limits, and maintenance procedures to meet the specific requirements of each battery type.

They are usually equipped with thermal sensors to monitor temperature, ensuring that the batteries and the controller maintain work temperatures. In cases of extreme battery load, where the batteries may overheat, installed external temperature sensors can be used to monitor the battery temperature more accurately, ensuring proper functioning and avoiding overheating. Usually, the charge controllers are equipped with some sort of display that shows critical parameters like battery voltage, state of charge, and both charge and load currents and allows tracking of the performance of the whole system by the user.

The charging process consists of four primary stages: bulk, trickle, absorption phase, and equalization charge. The bulking phase involves charging the battery with the maximum current generated by solar panels (MPP). As the battery charges, its voltage increases in line with its state of charge. During the Absorption phase, when the battery reaches its gassing voltage, the charge current is gradually reduced to avoid overcharging. The charge controller maintains the voltage until the battery reaches its maximum charge voltage. In the Trickle phase, when the charging current drops below a certain threshold, the controller detects the battery is fully charged and switches to trickle charging. At this point, small current pulses are used to keep the battery at full capacity. The controller shows the battery's state of charge, and temperature compensation could be used to adjust the charging voltage based on the battery's temperature. The last stage is the equalization charge, which is used to balance the battery by a short overcharge to balance the state of charge of all cells. This helps prevent problems such as sulphation or uneven wear and tear on individual cells. To ensure the safety of the system during this phase, the charge controllers are equipped with the above-

mentioned protection mechanisms to limit the power supplied by the solar panels and the load draw, if necessary, and to prevent overcharging. All four described phases can be seen in Figure 14.

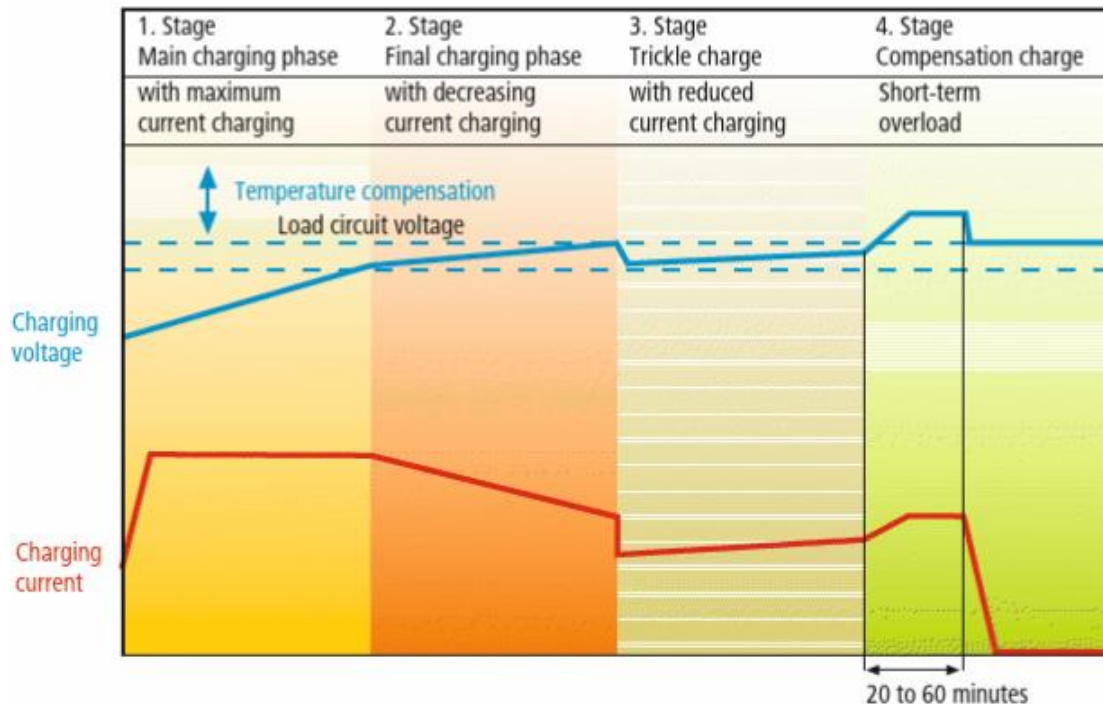


Figure 14: Graph of charging cycle of battery, source: Deutsche Gesellschaft für Sonnenenergie (2013)

2.3.5.1 Series Controller

This type of controller uses a series configuration of relays or semiconductor switches, usually MOSFETs, for switching off a charging current flow into the battery storage. Information on the operation and characteristics of these controllers is provided by Deutsche Gesellschaft für Sonnenenergie (2013, p. 165). Charging resumes when the voltage falls below a defined threshold. The solar panels must stay connected in order for the actuator to function reliably in the event that the battery is completely depleted. Previously, it was believed that series controllers had large power losses; however, the use of power semiconductors has decreased these losses. The power switch in a linear series controller does not turn off until the maximum voltage is reached. The series controllers have the benefit of being able to be connected to power sources without short-circuit protection. Furthermore, series controllers improve system efficiency and reliability by lowering the possibility of hot spots forming in solar modules, in contrast to shunt controllers. efficiency.

2.3.5.2 Shunt Controller

Their primary function is to protect batteries from overcharging by diverting excess energy away from the battery once it is fully charged. This operating principle, along with other characteristics of shunt controllers, is described in detail by Deutsche Gesellschaft für Sonnenenergie (2013, p. 166). Instead of interrupting the current flow, as series controllers do, shunt controllers "shunt" or redirect the surplus energy to an alternate path, such as a dump load or resistor, where it can be safely dissipated. As the controller continuously monitors a battery voltage status and activates when the voltage reaches a defined value. They are very cost-effective and relatively simple in their design; on the other hand, the efficiency is not that good compared to other controllers, as the produced energy is being converted purposely into heat. Shunt controllers are widely considered as the safest option in terms of cost.

2.3.5.3 PWM Controller

Building on the explanation provided by Heltsley (2023), PWM controllers regulate the power generated by solar panels by rapidly switching the current on and off in high-frequency pulses, typically using MOSFETs. This process prevents overheating and gassing of the battery while creating rectangular wave signals with a variable duty cycle, enabling precise control of the power delivered to the battery. Unlike MPP (Maximum Power Point) controllers, PWM controllers do not optimize the solar array's power output. As the battery approaches full charge, the controller gradually reduces the pulse width, lowering the charge current until the battery is fully charged. Although PWM controllers are less efficient than MPP controllers, they are more affordable and are commonly used in smaller systems. Additionally, they can help recover some lost battery capacity.

2.3.5.4 MPP Charge Controller

Based on the description provided by Deutsche Gesellschaft für Sonnenenergie (2013, p. 166), MPP (Maximum Power Point) charge controllers are specifically designed to maximize the efficiency of photovoltaic systems by ensuring the solar panels operate at their maximum power point, regardless of the battery's voltage. The battery voltage frequently determines the operating point on the voltage/current curve in conventional systems with series or shunt charge controllers, which can hinder the solar array from operating at its most efficient level.

This mismatch can result in power losses ranging from 10% to 40%, depending on factors like battery voltage, sunlight intensity, and temperature. To address this issue, an MPP tracker is used. This device includes a regulated DC-DC converter that dynamically adjusts the operating parameters of the solar array to extract the maximum available power. The MPP tracker periodically (e.g., every five minutes) scans the current-voltage characteristics of the PV array to identify the maximum power point. The DC-DC converter then adjusts the output to draw optimal power from the array while matching the voltage required to charge the battery.

2.3.6 Solar Panel Mounting Structures

2.3.6.1 Roof-mounted systems

In the context of small-scale photovoltaic deployment in the Czech Republic, roof-mounted systems remain a predominant choice, as they can be integrated directly into existing building structures. The construction principles and material solutions for these systems are outlined by Bolt (2024). A roof-mounted system basically uses the existing roof of the building on which the mounting structure is placed. These structures are designed for durability and corrosion resistance, with options including galvanized steel, stainless steel, and aluminum, which offer a lightweight and durable alternative. The type of structure depends on the type of roof, which can be either gable or flat (horizontal). Depending on the type of roof, a suitable construction solution is then chosen. For flat roofs, more complex structures are often used, which are slightly similar to ground-mounted systems. For gable roofs, special rails are used, which are laid horizontally to the roof. Clamps are then attached to these rails to hold the panel securely attached to the system structure. This structure securely holds the solar panels in place because the rails are attached to hooks, which connect the structure to the truss of a building. These hooks are typically secured to the building truss using bolts or screws. While this method generally follows the angle of the existing gable roof, flat roofs offer more flexibility, allowing for the selection of an optimal angle to maximize sunlight exposure on the mounted panels. Before installation, however, it is important to check whether the roof or roof truss is capable of supporting the weight of the solar panels and the whole mounting structure.



Figure 15: Gable roof-mounted photovoltaic system, source Fischer A. (2022)



Figure 16: Flat roof-mounted photovoltaic system, source: Hněvkovský J. (2023)

2.3.6.2 Ground-mounted systems

Jude (2024) provides a comprehensive overview of ground-mounted photovoltaic systems, which represent another common method of installing solar panels, particularly when rooftop space is limited or when optimal sun exposure is required. Ground-mounted systems are ideal for large-scale solar installations, such as those found in solar farms or for residential properties with ample land. The main components of a ground-mounted system include the mounting frame and support posts. The system is typically secured to the ground using metal or concrete supports, which are either driven directly into the soil or anchored with concrete footings.

The three main types of ground-mounted systems are standard ground mounts, pole-mounted systems, and solar carports. Standard ground mounts use steel frameworks fixed to the soil. Pole-mounted systems involve panels mounted on vertical poles, offering flexibility in panel tilt and height adjustment, especially on uneven terrain. Solar carports support panels above parking areas, providing both energy generation and shelter. Ground-mounted systems offer advantages like optimal sun exposure and easier maintenance compared to rooftop setups. However, they require more space and can be more costly to install. Despite these challenges, ground-mounted systems are efficient and scalable, making them suitable for a variety of applications.



Figure 17: Ground-mounted photovoltaic system, source: Solarise solar (2025)

2.3.6.3 Solar roof

There are two ways to create a solar roof: one involves the use of solar tiles, and the other utilizes standard photovoltaic panels laid as a complete roofing system, which is a less conventional but emerging method. As noted by Fotovia (2024), solar roof tiles integrate photovoltaic cells directly into the tiles, providing both roof protection and energy production. Their main advantage is their aesthetic appeal, as they are more visually pleasing than traditional solar panels. However, they are more expensive than conventional panels, with costs ranging from CZK 12,000 to CZK 24,000 per square meter. Additionally, their energy efficiency is slightly lower, as optimal panel performance is influenced by the roof's orientation and also tilt. In the event of damage, individual tiles can be replaced with minimal disruption to the system.



Figure 18: Roff solar tiles, source: Fotovia (2024)

The second option is to use standard solar panels as roofing, which is an innovative approach that combines energy production with the structural function of a roof. Although this concept is not yet widespread, it is becoming increasingly common, especially in the form of carports, pergolas, gazebos, and other backyard or auxiliary structures. These solar canopies provide usable space underneath while generating electricity, making them a suitable alternative for properties where rooftop installation is impractical or undesired (Hyder, 2025). A key technical challenge is ensuring proper sealing between the panels. Since solar panels are mounted with small gaps or elevated off the surface, water can penetrate through these spaces if the installation is not carefully designed. This may lead to moisture ingress under the structure, potentially reducing its lifespan. To address this, pre-engineered canopy systems and sealing components are often used to maintain both structural integrity and waterproofing.



Figure 19: Solar Canopy, source: Shelden (2024)

2.4 Concept of connection of a photovoltaic power plant

2.4.1 Grid connection

A grid-connected system is connected to the external power grid and allows the electricity generated by the system to be sent to the grid but also to draw power from the external grid in the event of a local power shortage. Owners are often entitled to earn money for the excess energy supplied to the grid through net metering. If solar energy production is insufficient, for example, at night or in cloudy weather, the grid serves as a backup energy source. Systems implemented in a grid connection must always have output parameters in line with the external power grid. Fortunately, this is ensured by the inverter, which is tasked with maintaining the parameters set by the grid (Pagaria, 2022). In the event of large grid overflows that could cause a grid collapse, the external grid operator can disconnect the photovoltaic system from the grid in this topology or disable the overflows.

2.4.2 Islanded connection

The following overview of island systems is based on the explanation provided by the PVcase Team (2023). This type of system operates independently of the external power grid. In this system, the generated energy is stored in batteries, which provide power when needed. The island topology is particularly used and useful in areas where there is no available or reliable external power grid. The primary drawback of the island topology is the potential for power shortages during extended periods of adverse weather and due to inadequate sunlight.

In such situations, the internal system would rely solely on batteries, which may lack the capacity to sustain the internal grid for an extended time duration. This problem highlights the importance of designing any photovoltaic system to have sufficient battery storage. To ensure this, it is essential to first assess and determine the daily energy needs of the house or facility, as this will determine the appropriate battery capacity. The total daily energy demand is then divided by the number of hours of peak sunshine in the area, and the required system size is calculated. Despite higher initial costs, mainly due to the need for battery storage, islanded solar systems provide long-term benefits in terms of independence as they do not use an external power grid.

2.4.3 Hybrid connection

Novergy (2024) outlines that a hybrid connection of a photovoltaic power plant integrates both islanded and grid-connected modes into a single system. The system is physically connected to the external grid while also including battery storage that enables it to operate independently when needed. The electrical energy generated by the solar panels primarily powers the internal grid, such as a household, while any surplus energy is stored in the battery. Once the batteries are fully charged, excess energy is fed into the external grid. In times of low solar production, the system can draw power either from the batteries or from the external power grid to ensure an uninterrupted supply of electricity.

2.5 Photovoltaic energy in the Czech Republic

2.5.1 Legislation

As discussed by Isofen Energy (2009) and later expanded by Kubíska & Prieložný (2023), the Czech Republic made a significant stride in promoting the generation of electrical energy through photovoltaics. This development has been largely driven by the implementation of strong legislative support. Act No. 180/2005 Coll. on the Promotion of Electricity Production from Renewable Energy Sources, in particular, established the legal framework required to encourage the use of solar technologies. One of the main purposes of this law was to integrate renewable energy sources into the energy system of the Czech Republic and thus continue to support their development.

Because of this law, which was supported by substantial government subsidies and fixed feed-in tariffs, the photovoltaic capacity increased quickly, especially between 2009 and

2011, by providing a stable and alluring environment for solar developers and investors. This law laid a solid foundation for the era of photovoltaics in the Czech Republic because, under this law, feed-in tariffs were set for each commissioning year in a way that aimed to ensure a payback period of 15 years, with the guaranteed price fixed for already commissioned installations, as it was described in Chapter 3.1. No. 165/2012 Coll.'s main purpose was to modify some previous conditions, but also to introduce more flexible and supporting mechanisms that are more responsive to market changes and technological progress. This law also regulated some conditions for operators of photovoltaic power plants and introduced a levy on the electricity produced in this way. The main objective was to regulate and balance financial support for solar power plants. Another change was brought by Act No. 131/2015 Coll. and Act No. 369/2016 Coll., which modified some existing obligations but mainly added administrative steps in the form of reports for owners and investors. Act No. 131/2015 Coll. focused more on the rules and obligations related to the connection of new power plants. Act No. 225/2017 Coll., then amended the feed-in tariffs and green bonuses associated with photovoltaic power plants.

Janíček, Reichmann, Huťan, & Samek (2024) notes that the implementation of a photovoltaic power plant in the Czech Republic requires fulfilling several administrative conditions. One of the key requirements is obtaining a license to operate the PV system, though this obligation does not apply to off-grid installations or systems with an installed capacity of up to 50 kWp. Another common requirement is a building permit issued by the competent authority. However, similar to the licensing exemption, this permit is not necessary for plants under 50 kWp unless the installation significantly affects the structural integrity of the building, involves a heritage-protected structure, or alters the building's intended use. These requirements are defined under Building Act No. 183/2006 Coll.

2.5.2 Nová Zelená úsporám

Based on information provided by the Státní fond životního prostředí ČR (2025, pp. 23-24), the "Nová zelená úsporám" (New Green Savings) subsidy fund was established, which also focused on supporting newly built photovoltaic power plants for family houses and industry. The amount of this subsidy has already been amended several times and according to the latest data (21.1.2025), it is possible to receive a subsidy of up to CZK 100,000 per installed system. A subsidy of up to CZK 140,000 may also be granted, provided that the installed power

plant is connected to a system for sharing the energy produced at the same time as the "smart device" function. The "smart device" function can be understood as meaning that the PV system can be disconnected remotely by the grid distributor, also limiting the overflows to the grid and the system must smartly manage the produced energy (charging the batteries, heating the house and heating the water with the excess energy, etc.). One of the basic conditions is that the subsidy can be applied for from a minimum installed capacity of 2 kWp. To qualify for this state subsidy, some other specific conditions must be met, including that the technical specifications are met (MPPT technology, minimum 95% efficiency of the hybrid inverter), the building must be used for permanent housing, and the building must be connected to the external electricity grid. The energy performance of the building also plays a role in the final amount of the subsidy, as does the size of the installed system in relation to the energy savings achieved.

2.5.3 The Green Bonus in the Czech Republic

As described by Solární Novinky (2022), the Green Bonus was another way to encourage the development of photovoltaic power plants in the Czech Republic, and it was introduced under Act No. 180/2005 Coll. Unlike fixed feed-in tariffs, the Green Bonus was a premium paid to producers for each MWh of electricity delivered to the grid. Its amount was not subject to year-on-year guarantees and could fluctuate depending on market conditions. As the cost of photovoltaic systems declined rapidly after 2010, the Energy Regulatory Office (ERO) gradually reduced the Green Bonus. While some PV owners considered these changes economically unfavorable, they were legally anticipated and aligned with the original legislative framework. In 2023, the Green Bonus scheme formally ended for PV systems commissioned between 2011 and 2013. The significant reduction in support has been criticized in particular by investors. These adjustments were part of broader efforts to align support mechanisms with current market conditions.

2.5.4 Development of photovoltaic feed-in tariffs

The introduction of feed-in tariffs also took place in 2005, and their evolution was quite similar to the green bonus. In 2008, the maximum guaranteed feed-in price was up to CZK 18 876 for each MWh produced, regardless of the size or capacity of the power plant. However, this support decreased with the number of new installations. By 2011, feed-in tariffs for new

installations had decreased significantly, depending on the capacity of the plant. In addition, in 2010, the Czech government amended the legislation to allow for reductions in feed-in tariffs greater than the previously permitted 5% per year, in response to the rapid decline in photovoltaic system costs. Act No. 165/2012 Coll. effectively ended support for new PV plants from 2011, with a formal stop to feed-in tariffs for new installations from 2014. From 2013, the green bonus became market-based and hourly, meaning its real value could fluctuate significantly, and in some cases, even turn negative. Thus, published tariff rates after this period were only indicative. This step was crucial for the transformation of the Czech solar market, but brought a number of problems for smaller electricity producers who were unsure of the return on their investment. The evolution of green bonuses and feed-in tariffs is clearly illustrated in Figure 20, which also includes period-specific prices.

	Podporovaný druh energie	Datum uvedení výroby do provozu		Instalovaný výkon výroby [kW _e]		Výkupní ceny [Kč/MWh]	Zelené bonusy [Kč/MWh]
		od	do	od	do (včetně)		
ř./sl.	a	b	c	d	e	f	g
500	Výroba elektřiny využitím slunečního záření	-	31.12.2005	-	-	9 222	8 258
502		01.01.2008	31.12.2008	-	-	18 876	17 912
503		01.01.2009	31.12.2009	0	30	17 710	16 746
504		01.01.2009	31.12.2009	30	-	17 579	16 615
505		01.01.2010	31.12.2010	0	30	16 495	15 531
506		01.01.2010	31.12.2010	30	-	16 362	15 398
507		01.01.2011	31.12.2011	0	30	9 896	8 932
508		01.01.2011	31.12.2011	30	100	7 788	6 825
509		01.01.2011	31.12.2011	100	-	7 257	6 293
510		01.01.2012	31.12.2012	0	30	7 969	7 006
511		01.01.2013	30.06.2013	0	5	4 325	3 361
512		01.01.2013	30.06.2013	5	30	3 590	2 627
513		01.07.2013	31.12.2013	0	5	3 791	2 827
514		01.07.2013	31.12.2013	5	30	3 085	2 121

Figure 20: The Green Bonus and feed-in tariffs, source: TZB-info (2024)

3 Aim of Diploma Thesis

The aim of this diploma thesis is to evaluate and compare the real-world performance of two photovoltaic (PV) power plants with distinct installed capacities and technical configurations. The objective is to assess operational parameters and key performance indicators, such as energy efficiency, performance ratio, self-consumption, and grid dependency, based on comprehensive monitoring data collected over a two-year period. The thesis focuses on identifying how system-specific factors, such as panel orientation, tilt angle, component selection, and local environmental conditions, influence energy production under actual operating conditions.

Key performance indicators (KPIs) are calculated and interpreted in accordance with established international methodologies, particularly the ČSN EN IEC 61724-1 ed. 2 standard and other sources. Additionally, real production data are compared with theoretical values generated by the PVGIS simulation tool, enabling an evaluation of predictive accuracy and the identification of discrepancies attributable to weather variability or system-specific limitations.

This comparative approach allows for a nuanced understanding of system behavior and highlights opportunities for operational optimization. While the primary focus of the thesis is analytical and does not aim to offer prescriptive solutions, several practical recommendations are proposed where performance trends and observed inefficiencies clearly justify them.

4 Methodology and Materials

The practical part of this thesis focuses on evaluating the performance and efficiency of two monitored photovoltaic power plants, Power Plant A and Power Plant B, situated in close geographic proximity. The primary objective of this section is to assess and compare the energy production behavior of each plant over the course of two years (2023 and 2024), with a particular focus on identifying the key technical and environmental factors that influence their operational characteristics and performance indicators.

The two monitored systems differ not only in installed capacity but also in their technical design, including panel types, orientation, tilt angle, and slightly different local irradiation conditions, all of which significantly affect system performance. A comparative analytical approach is applied, based on real monitored data, to evaluate the systems under actual operating conditions. The analysis tries to examine how these technical configurations affect daily and seasonal variations in energy yield and system efficiency.

The following methodology is structured into four main sections: data collection, technical part, key performance indicators, and operational parameters. Each section outlines the procedures and tools used to analyze system performance and interpret the measured data under real operating conditions. In the final step, the monitored values were compared with PVGIS-based simulations to assess their accuracy and identify any real-world deviations.

4.1 Data collection

Both Power Plant A and Power Plant B are equipped with SolarEdge inverters that integrate advanced monitoring capabilities for continuous tracking of system parameters. These inverters collect detailed performance data, including AC power output, exported energy to the grid, system efficiency, and additional electrical characteristics. For the purposes of this thesis, retrospective datasets were extracted via the SolarEdge monitoring platform, allowing for long-term performance analysis based on historical energy production and consumption records. The acquired data were used to quantify operational efficiency and temporal trends under real-world conditions. Due to different commissioning dates, Power Plant A having been operational earlier, a temporal offset exists between the datasets. This discrepancy was carefully accounted for during the analytical evaluation to maintain consistency and ensure the validity of comparative results.

4.2 Technical part

The methodology for evaluating the performance of Power Plant A and Power Plant B is based on key performance indicators (KPIs) aligned with industry standards and supported by critical operational parameters. The analysis and calculations in this section follow the guidelines established by the ČSN EN IEC 61724-1 ed. 2 standard, including the selected equations provided therein. The evaluation focuses on essential metrics such as energy production, efficiency, and overall system sustainability. Solar irradiation data, crucial for understanding the energy potential of each site, were sourced from reputable tools such as the Solar Atlas and PVGIS, which provide detailed insights into the solar resources available at each power plant's location. The technical evaluation also considers the overall configuration and performance of the system, detailing the types of solar panels, inverters, and other key components used in the power plants. This comprehensive approach enables a thorough comparison of the two power plants based on their design, operational characteristics, and the factors influencing their energy production and efficiency.

4.3 Key Performance Indicators and Crucial Operational Parameters

The assessment of the performance and operational efficiency of a photovoltaic (PV) power plant requires the monitoring and evaluation of both Key Performance Indicators (KPIs) and Crucial Operational Parameters. These indicators and parameters provide a comprehensive understanding of the system's efficiency, capacity, and operational conditions, ensuring effective management and optimization of the power plant's performance.

4.3.1.1 Key Performance Indicators (KPIs)

The KPIs are used to evaluate the performance and efficiency of a PV power plant in achieving its operational goals. They focus on the plant's actual performance. The following KPIs are critical for assessing the overall performance of the system:

Total Energy Produced from the Sun – represents the total electrical energy generated by the PV system over a specific period.

Total Energy Drawn from the Grid – This value quantifies the amount of energy imported from the grid when solar production is insufficient to meet demand.

Total Energy Sold to the Grid – This metric tracks the excess energy exported to the grid, providing insights into potential revenue streams and the plant's contribution to the overall energy mix.

Total Energy Produced for On-Site Consumption – It evaluates how much of the PV-generated energy is used on-site instead of being exported to the grid.

Energy Exports – is the percentage of the total energy produced by a photovoltaic power plant that is fed back into the external grid rather than being consumed on-site. It reflects how much of the generated energy is surplus to the system's immediate needs and can contribute to the energy supply of others.

$$\text{Energy Exports} = \frac{\text{Energy Sold to the Grid}}{\text{Total Energy Produced}} \times 100 \text{ [\%]} \quad (12)$$

This formula is introduced in this work based on a logical evaluation of system energy flows.

Overall Self-Sufficiency – The proportion of total energy consumption that is covered by the on-site production of the photovoltaic system. A higher value indicates reduced reliance on the external grid and greater energy independence.

$$\text{Overall Self-Sufficiency} = \frac{\text{Total Energy Produced for On-Site Consumption}}{\text{Total Energy Consumed}} \times 100 \text{ [\%]} \quad (13)$$

This indicator is defined in this thesis for the purpose of evaluating energy independence.

Utilization of own Production – Indicates the share of the total generated solar energy that is directly consumed on-site. It helps evaluate how efficiently the system uses its own generation before exporting excess energy to the grid.

$$\text{Utilization of own production} = \frac{\text{Total Energy consumed On-Site}}{\text{Total Energy produced}} \times 100 \text{ [\%]} \quad (14)$$

Proposed to quantify how much of the generated solar energy is consumed on-site before being exported.

Grid Dependency - measures the level of reliance on the external grid for meeting the energy needs of the plant. A lower grid dependency indicates better independence of the power plant.

$$\text{Grid dependency} = \frac{\text{Total Energy drawn from the grid}}{\text{Total Energy consumed}} \times 100 \text{ [\%]} \quad (15)$$

Presented here to measure reliance on external energy sources in relation to total system demand.

Performance Ratio – It compares the actual energy output of the system to its theoretical maximum potential output, based on the system's installed capacity and the average solar irradiation for the region. PR provides an assessment of how efficiently a solar system converts sunlight into electricity by comparing the actual energy produced to the theoretical energy output, which is calculated using the installed capacity and average annual solar irradiation. By the performance ratio, the efficiency with which the system converts solar energy into usable electrical energy, while accounting for real-world factors, can be determined.

$$\text{Performance Ratio} = \frac{\text{Actual energy output}}{\text{Installed capacity} \times \text{Average solar irradiance}} [-] \quad (16)$$

Source: ČSN EN IEC 61724-1 ed. 2:2022, Photovoltaic system performance

Specific Yield – It determines the amount of energy produced per unit of installed capacity over a given period. It measures how efficiently the system converts its installed kWp (kilowatt-peak) into actual energy (kWh) throughout the year. It is expressed in kWh per kWp per year.

$$\text{Specific Yield} = \frac{\text{Annual Energy Production}}{\text{installed Capacity}} [\text{kWh/kWp/year}] \quad (17)$$

Source: ČSN EN IEC 61724-1 ed. 2:2022, Photovoltaic system performance

Capacity utilization Factor – used to assess the efficiency of a power generation system, representing the ratio of actual output to the maximum possible output over a specific period. It is calculated by dividing the total energy produced by the system during a time frame by the energy that would be produced if the system operated at its rated capacity continuously. This calculation was adopted from

$$\text{Capacity utilization factor} = \frac{\text{Annual Energy Production}}{\text{Rated Capacity} \times \text{Hours annually}} [-] \quad (18)$$

Source: IRENA (2021). Renewable Power Generation Costs in 2020; SMA Solar Technology AG (2011).

4.3.1.2 Crucial Operational Parameters

These include technical and environmental factors that influence the PV plant's performance. They are not direct performance metrics but rather parameters that impact the KPI values section. The key metrics and operational characteristics are presented and categorized accordingly.

System Rated Capacity: It represents the total output the system can generate when exposed to ideal conditions, such as full sunlight with no shading or environmental losses. This value is determined by the combined rated power of all installed solar panels and serves as an indication of the system's design capabilities. It is given in kilowatts peaks (kWp)

Panel area – The area of each individual panel used in the monitored power plants is taken into account, allowing for an evaluation of space utilization and its impact on energy production.

$$\text{Panel area} = \text{height} \times \text{width} [\text{m}^2] \quad (19)$$

Standard geometric formula for surface area.

Total area of panels – The total installed panel area is calculated by multiplying the number of panels in the system by the area of each panel. This metric provides an overall indication of the physical size of the installation.

$$\text{Total area of panels} = \text{Panel area} \times \text{number of panels} [\text{m}^2] \quad (20)$$

Calculated as the product of unit panel area and total number of modules.

Panel Area per 1 kWp – This formula indicates the required area per kilowatt-peak (kWp) of installed capacity. It provides insights into system design efficiency and space use in terms of power generation. To assess the spatial efficiency of the photovoltaic system, the Panel Area per 1 kWp metric was calculated using the formula:

$$\text{Panel area per 1 kWp} = \frac{\text{Total panel Area}}{\text{Installed capacity}} [\text{m}^2/\text{kWp}] \quad (21)$$

Source: Raman, Madhumathi and DARCY, Michael. Solar Panel Calculator. Omni Calculator, 2023.

Specific Area of PV per 1 kWh produced annually – this parameter calculates how much panel area is needed to generate one kilowatt-hour (kWh) of electricity. It helps assess the energy production efficiency of each power plant by correlating the energy output with the required panel area.

$$\text{Specific Area of PV per 1 kWh annually} = \frac{\text{Total panel Area}}{\text{Annual Energy Production}} [\text{m}^2/\text{kWh}] \quad (22)$$

This formula was introduced by the author to enable a spatial evaluation of energy yield based on commonly accepted photovoltaic system parameters

Panel Area Needed to Produce 1 kWh/kWp Annually – This calculation determines the amount of panel area required to produce one kWh of electricity per kWp of installed capacity over a year.

$$\text{Panel Area per 1 kWh/kWp} = \frac{\text{Installed capacity} \times \text{Total panel Area}}{\text{Annual Energy Production}} \quad [\text{m}^2/\text{kWh/kWp}] \quad (23)$$

As no direct equivalent was found in the literature, this formula was developed to assess long-term spatial performance efficiency in relation to installed capacity.

Average Panel Efficiency under Actual Operating Conditions – This formula provides a realistic measure of how effectively the solar system operates in its specific environmental context, considering real-world conditions and not just theoretical maximums. By calculating this efficiency, it is possible to evaluate the overall performance of a solar energy system and understand how much of the incident solar energy is actually converted into electrical energy by the panels.

$$\text{Average Panel Efficiency} = \frac{\text{Actual energy production}}{\text{Total panel area} \times \text{Solar irradiance}} \times 100 \quad [\%] \quad (24)$$

Source: Svarc, Jason. Most Efficient Solar Panels 2025. Clean Energy Reviews, 2025.

Weighted Average Panel Efficiency – It calculates the overall efficiency of a solar system by considering both the efficiency and the area of each panel type. It accounts for variations in panel efficiency by weighting each panel's contribution based on its area, providing a more accurate reflection of the system's performance compared to a simple average. By considering both the efficiency of each panel and its area, it provides a more accurate and holistic measure of the system's real-world performance.

$$\text{Weighted Average Panel Efficiency} = \frac{\sum_{i=1}^n \text{Efficiency of panels}_i \times \text{Area of panels}_i}{\sum_{i=1}^n \text{Area of panels}_i} \quad (25)$$

This formula was proposed by the author based on the general principle of a weighted average.

4.4 Photovoltaic power plants

4.4.1 Power Plant A

Power Plant A is located in the village of Sentice, in the Tišovsko region. The photovoltaic power plant was installed primarily to offset the high electricity costs resulting from significant grid consumption. The building, an industrial facility, houses a food processing plant that requires a stable indoor temperature. It features multiple cooling rooms and tanks,

all maintained by a central cooling system powered by a high-performance air conditioning compressor. Due to the substantial energy demand for cooling and consistently high overall consumption throughout the year, especially during the summer months, the investor decided to implement a photovoltaic system. By producing its own electricity, the facility minimizes its dependence on the grid while optimizing the use of solar energy. This arrangement is especially advantageous since the plant's retail and manufacturing operations occur during daylight hours, exactly when solar panels can generate electrical power. Consequently, the facility can utilize the energy it produces in real time, significantly reducing its reliance on external electricity sources.

The power plant was constructed in two phases. In the first phase, 50 DAH 400 Wp (HCP78X9) panels were installed, forming the initial foundation of the system. The second phase, completed in 2022, expanded the installation with 44 newer 450 Wp SSY Jinko Solar panels (JKM450M). These newer panels were integrated into two additional strings, bringing the total system to four strings and a combined installed capacity of 39.8 kWp. All panels are mounted at a 25° tilt.

Table 1: Photovoltaic panels of power plant A, source: author

	SSY Jinko Solar JKM450M 60HL4	DAH Solar HCP78X9- 400W
Type	P-type Mono-crystalline	Mono-crystalline
Number of cells	120	156
Dimensions	1903x1134x30 mm	2008x1002x30 mm
Weight	24.2 kg	24.3 kg
Frame	Anodized Aluminium Alloy	T6 Frame (aluminium alloy)
Module Efficiency STC	20.85 %	19.88 %
Maximum power (Pmax) - NOCT	335 Wp	302 Wp
Maximum power (Pmax) - STC	450 Wp	400 Wp

As previously mentioned, the power plant consists of a total of four strings, with two oriented to the southwest, azimuth 130°, and the other two oriented on the northwest at azimuth -50°. The installation of this updated photovoltaic system took place in the autumn of 2022, followed by subsequent commissioning and startup. As a result, the data utilized in this thesis is derived from readings and measurements collected starting January 1, 2023, which accurately reflect the performance of the power plant under its new configuration. The Jinko Solar JKM450M panels, along with the previously installed DAH solar panels, were integrated with SolarEdge S500-1RN4MRM optimizers as part of the upgrade. This retrofit ensured that both the new and existing panels operate under the same optimization system, allowing for better performance, more precise power management, and improved reliability of the entire power plant. The system does not include any form of battery storage; therefore, any surplus energy generated is fed into the grid and sold.

Table 2: Inverter properties power plant A, source: author

SolarEdge SE17K 3-phase inverter	
Rated AC power output	17,000 VA
Maximum Continuous Output Current per phase	26 A
Ac Frequency	50/60 +- 5 Hz
Maximum DC Power	22,950 W
Maximum Inverter Efficiency	98 %
Maximum DC Input Voltage	900 Vdc
Maximum Input Current	23 Adc
Nominal DC Input Voltage	750 Vdc
Supported Communication Interfaces	TS485, Ethernet, Wi-Fi, Built-in GSM,

The photovoltaic power plant is installed on the roof of the building, with all cabling from the solar strings routed through a UV-resistant conduit, which is securely affixed to the building's facade. The cables connecting the solar strings to the DC distribution box have a cross-sectional area of 6 mm². Cables lead from the DC distribution box, which houses the necessary fuses for overcurrent protection, to two SolarEdge SE17K three-phase inverters. Each inverter is responsible for two strings. From the inverters, the cables then connect to the

internal AC network via CYKY-J 4x25mm cables, linking the inverters to the AC distribution box. Figure 21 shows the whole described circuit diagram of photovoltaic power plant A.

The costs for implementing the first phase amounted to 648,965 CZK, excluding VAT. The total expense for installing the two additional strings was 689,875.50 CZK, excluding VAT. Therefore, the overall project cost reached 1,338,840.50 CZK, excluding VAT. Both figures were verified and provided by the investor and the owner of the photovoltaic power plant.

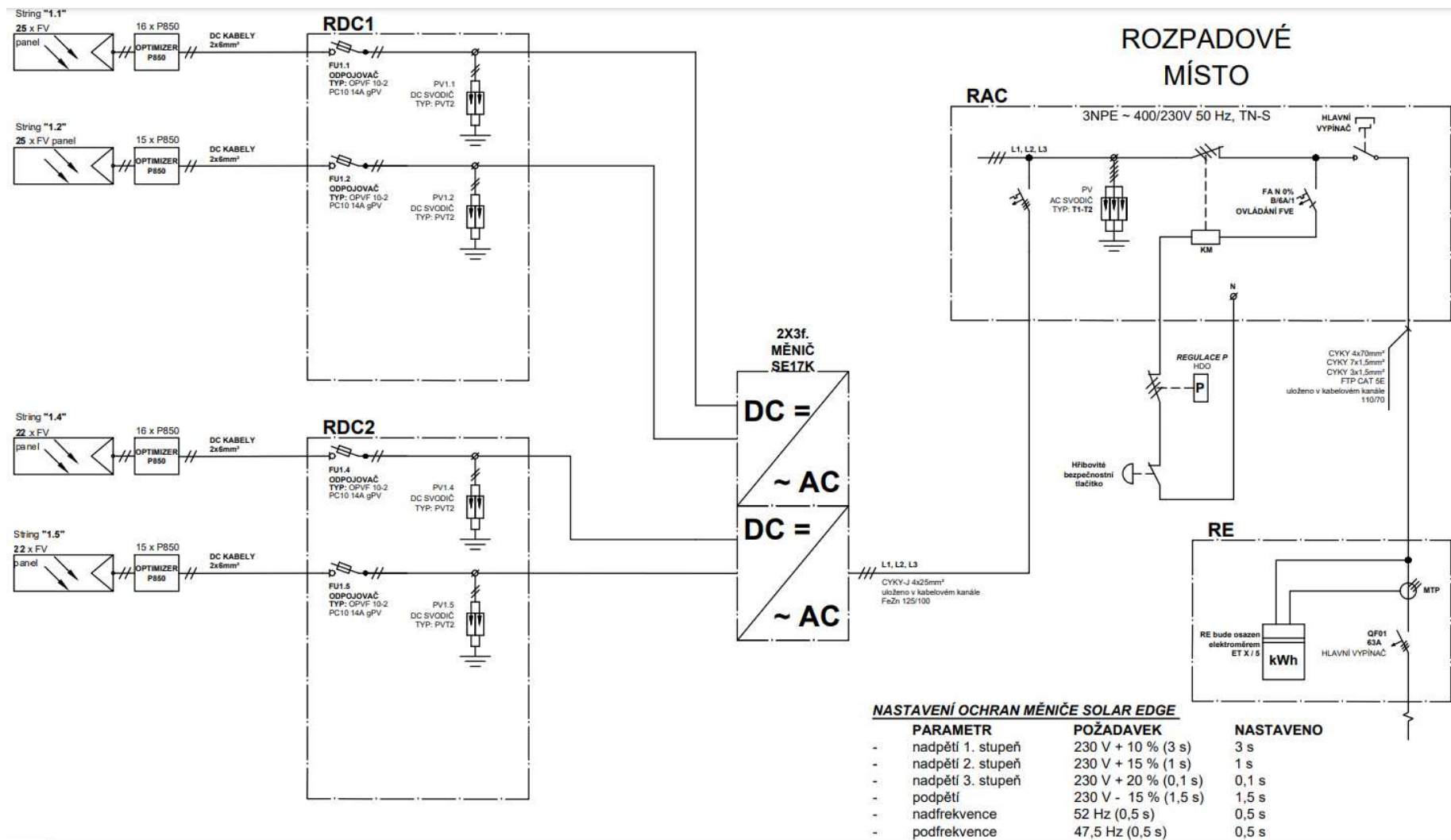


Figure 21: Circuit diagram of photovoltaic power plant A, source: Solelektro s.r.o., internal documentation (2022)

4.4.2 Power Plant B

Power Plant B is located in approximately the same area as Power Plant A. It is located in the village of Čebín, which is about 2 km away. Like Power Station A, it is an industrial facility, but this site is focused on metal processing and the production of various metallic components. The industrial building houses metal processing machines like TOS RFR 300/8, PUMA 700 XLY 2, DOOSAN VCF 850 SLR, and others (KIPO, 2021). The facility operates primarily during the day, as it runs on a two-shift schedule. For this reason, no battery storage system has been installed since all the electricity generated is either consumed directly or any surplus is fed into the grid and sold. The power plant officially began its operation on March 24, 2023.

The power plant is installed on the roof of the factory hall and consists of 366 solar panels with a total installed capacity of 139.44 kWp. The system consists of eight strings, with four oriented to the southeast, azimuth of -63° and four facing west, azimuth 116° . Each string is made up of 42 panels, manufactured by HT-SAAE, specifically the HT54-18X 415 Wp model, which features half-cut cell technology. This configuration ensures optimized energy production throughout the day, with the southeast-facing strings capturing the morning sunlight and the west-facing strings absorbing sunlight in the afternoon. The panels are installed at a 15° tilt. The half-cut cell technology improves efficiency by reducing losses due to shading and enhancing overall performance. The HT-SAAE HT54-18X panels are connected to SolarEdge P800 Worldwide 850 W optimizers. The panels are mounted on a self-ballasted aluminium structure, which ensures stability and durability without the need for roof penetration.

Each string is individually fused within a DC string box positioned on the side of the aluminum mounting structure, allowing for convenient access during maintenance or repair. The fuses inside the DC string box ensure safety by preventing potential overloads or short circuits and also safeguard the entire DC system. Since the power plant is situated on the roof, all cabling is routed through a UV-resistant conduit designed to endure significant temperature fluctuations and other conditions, including, as it was mentioned, UV radiation. The inverters are installed under a shelter located on the northern side of the building, offering protection while maintaining easy access for maintenance and monitoring of the system. For the DC part, 6 mm² solar cables were selected. From the DC distribution box, the cables lead

to the SolarEdge SE30K three-phase inverters, with one inverter acting as the master and the other three functioning as slaves. Collectively, these inverters provide a potential maximum output of 180 kW; however, this output is intentionally limited to 140 kW. The outputs from the inverters are directed at a dedicated distribution box. From this distribution box, the electrical power is conveyed to the main distribution box via CYKY-J 4x240 mm² cable.

Table 3: Panel properties of power plant B, source: author

HT-SAAE HT54-18X 415Wp	
Type	Mono-crystalline
Number of cells	108
Dimensions	1724x1134x30mm
Weight	21.0 kg
Frame	Anodized Aluminium Alloy
Module Efficiency	21.2%
Maximum power (Pmax) - NMOT	309Wp
Maximum power (Pmax) - STC	415Wp

Table 4: Inverter properties of power plant B, source: author

SolarEdge SE30K 3-phase inverter	
Rated AC power output	29,990 VA
Maximum Continuous Output Current per phase	43,5 A
Ac Frequency	50/60 +- 5 Hz
Maximum DC Power	45,000 W
Maximum Inverter Efficiency	98,3 %
Maximum DC Input Voltage	1000 Vdc
Maximum Input Current	43,5 Adc
Operating Voltage Range	680-1000 Vdc
Supported Communication Interfaces	RS485, Ethernet, Wi-Fi, Cellular,

All inverters are interconnected using UTP cables to facilitate both centralized control by the PLC unit and seamless communication between the master and slave inverters. The PLC

controller oversees and monitors the entire system, dynamically regulating the output of each inverter as needed to optimize performance. Additionally, it is designed to allow remote disconnection by the distributor if necessary, thereby enabling them to limit the supplied energy or completely disconnect the power plant to maintain grid stability and safety.

The total cost for the installation was 3,466,497.25 CZK, excluding VAT. This comprehensive investment covers all materials, equipment, and labour required to implement the photovoltaic system. Figures 22 and 23 collectively illustrate the complete electrical configuration of the system for Power Plant B. Figure 22 details the DC portion, showing the wiring from the photovoltaic panels through the DC distribution components, but it also includes the AC section after the inverters. Figure 23 further expands on the AC section, displaying the connections from the inverters to the AC distribution box and the main grid interface. Together, these figures provide a comprehensive overview.

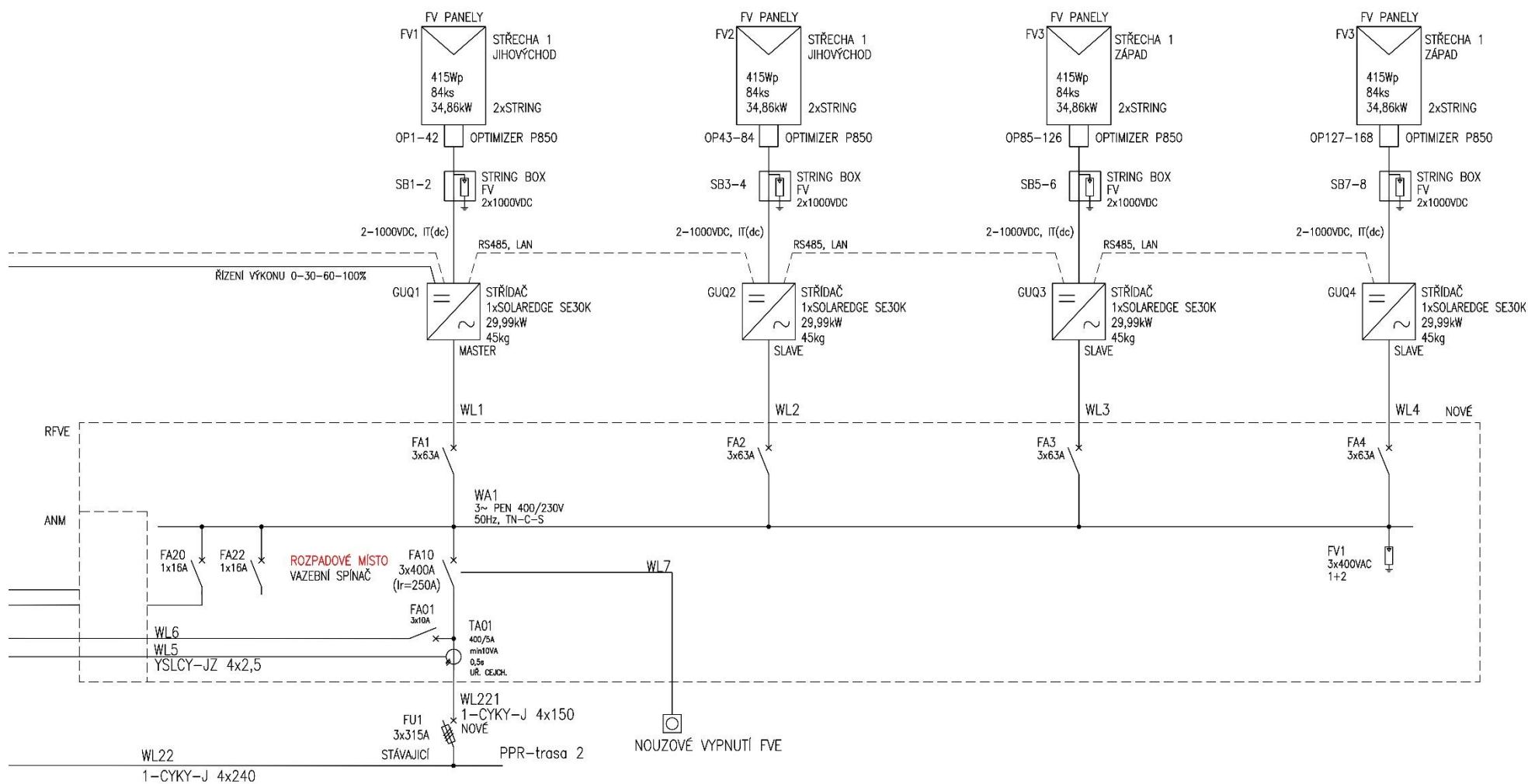
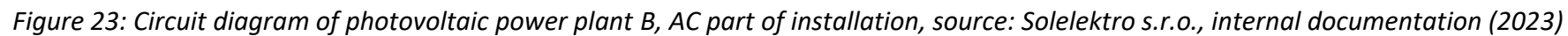


Figure 22: Circuit diagram of photovoltaic power plant B, DC part and inverters, source: Solelektro s.r.o., internal documentation (2023)

TS 22/0,4kV
Čebín – KIPO
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4.4.3 Solar irradiation in power plant locations

In Section 2.2.3 of this thesis, a solar irradiation map of the Czech Republic was presented. Using data from the Global Solar Atlas, the solar irradiation values for each power plant location were determined. Despite the fact that both power plants are situated in almost the same geographic area, minor differences in solar radiation data were observed. For Power Plant A, the Global Horizontal Irradiation (GHI) was found to be 1150.4 kWh/m² per year, while the Global Tilted Irradiation (GTI) per year was found to be 1313.4 kWh/m². The optimal tilt angle for PV panels at this location was determined to be 36 degrees. For Power Plant B, the GTI reached only 1277.4 kWh/m² per year, and the optimal tilt angle remained the same at 36 degrees. (Solargis, n.d.) The GTI differs because of the tilt of the mounted panels. Based on this comparison, it can be concluded that Power Plant A likely has a marginally better solar potential, as its panels are positioned closer to the optimal tilt angle.

5 Results and discussion

5.1 Analysis of Monitored Data and Utilization Efficiency

The following table presents key energy parameters recorded over the monitored operational period for both Power Plant A and Power Plant B. These parameters include totals of energy consumption, energy generated, energy drawn from the grid, energy sold to the grid, and energy produced for on-site consumption. Using Table 5, the given parameters can be displayed in a single, clear format, providing a comprehensive overview of energy production and its utilization. Based on this data, the overall energy self-sufficiency, own production utilization, energy export, and grid dependence were subsequently calculated.

Table 5: Analysis and Comparison of the Energy Balance Between Power Plants A and B

	<i>Power plant A system</i>	<i>Power plant B system</i>
Total energy consumption	110.214 MWh	422.799 MWh
Total energy made from the Sun	86.38 MWh	254.062 MWh
Total energy drawn from the grid	45.57 MWh	225.431 MWh
Total energy sold to the grid	21.736 MWh	56.694 MWh
Total Energy Produced for On-Site Consumption	64.648 MWh	197.368 MWh
Overall self-sufficiency	58.65 %	46.68 %
Utilization of own production	74.84 %	77.68 %
Energy exports	25.16 %	22.32 %
Grid dependency	41.35%	53.32 %

Power Plant B facility exhibits a much higher total energy demand, consuming 422.799 MWh, whereas Power Plant A facility requires considerably less energy, with a total

consumption of 110.214 MWh. This disparity stems from the unique operational characteristics of each facility. Power Plant B supplies energy to a high-demand industrial site, requiring a continuous and substantial electricity supply. In contrast, Power Plant A facility serves a smaller-scale operation with lower overall consumption, where a central cooling system is a primary energy driver.

Power Plant B produced a total of 254.062 MWh, which is supported by its larger installed capacity of 139.44 kWp. Notably, it also demonstrates a relatively high utilization rate, with 77.68% of the generated energy used on-site. Only 22.32% of the energy produced by Power Plant B is exported to the grid. In contrast, Power Plant A produced 86.38 MWh of energy, with 64.648 MWh, or 74.84%, being consumed on-site. 25.16% of Power Plant A's generated energy is sent to the grid. This slightly higher export rate indicates that Power Plant A has a higher proportion of surplus energy relative to its consumption, which is sent to the external grid. Despite this surplus energy being exported, Power Plant A still manages to achieve a relatively high self-sufficiency rate of 58.65%, reflecting its ability to consume a significant portion of the energy it generates on-site.

Furthermore, the above-mentioned arguments are supported by the fact that Power Plant A's facility has a lower grid dependency, with only 41.35% of its energy needs being met by the external grid, compared to Power Plant B's facility, which has a grid dependency of 53.32%. This reflects the more self-sufficient nature of Power Plant A in meeting its energy needs from its own production. On the other hand, Power Plant B's higher energy consumption and reliance on the grid highlight the challenges associated with larger energy demands that cannot be entirely met by self-generated power. Although Power Plant B is more efficient in utilizing its own energy, its larger operational demand makes it heavily reliant on grid-supplied electricity to maintain stable operation. As a result, Power Plant B's total energy consumption is significantly higher than that of Power Plant A, reaching 352.953 MWh compared to 110.073 MWh.

5.2 Analysis of Energy Production and Consumption Over Time

5.2.1 Power Plant Site A

5.2.1.1 The Year 2023

The graph presented in Figure 24 illustrates the energy production and consumption patterns of the photovoltaic power plant throughout 2023, highlighting distinct seasonal variations that influence the system's ability to meet the facility's energy demands. A clear correlation between solar availability and system output can be observed, with production rising steadily from winter to summer before declining toward the end of the year

During the winter months, the photovoltaic power plant struggles to generate sufficient electricity to cover the facility's demand. The primary reasons for this limitation include the lower solar elevation angle, shorter daylight hours, and frequent cloud cover, all of which significantly reduce available solar energy. Additionally, the facility's energy consumption remains relatively low during this period, likely due to reduced demand for cooling systems, which are predominantly active during warmer months. Despite this lower consumption, the system's output is insufficient to meet demand, necessitating increased reliance on grid electricity

A shift occurs in early spring, as longer daylight hours and still cooler temperatures. By mid-March, the power plant began to meet the facility's energy demand, reaching daily production of up to 170 kWh. As shown in the graph, consumption starts to rise in mid-March, but the system continues to generate enough energy to cover demand, even producing occasional surpluses. During April and May, the photovoltaic system reaches near self-sufficiency, with excess energy frequently available for export to the grid. This trend reflects optimal photovoltaic performance during spring, where prolonged sunlight exposure and moderate module temperatures contribute to enhanced energy yields.

The system reaches its annual production peak in July, with a recorded maximum output of 290.32 kWh per day. During the summer months, energy production is generally sufficient to meet the facility's increasing electricity demand, primarily driven by the extensive use of the cooling system. However, despite the high solar irradiation, consumption gradually surpasses generation in late summer as cooling and other equipment remain consistently high. Additionally, the highest recorded daily energy consumption occurred on August 24,

2023, reaching 286.552, reflecting peak operational loads during the hottest period of the year. This shift signifies the impact of seasonal energy demand variations on overall system efficiency.

As autumn progresses, energy production declines more rapidly than demand, driven by reduced solar radiation and shorter daylight hours. By December, production dropped significantly, with a recorded system shutdown between December 2 and December 11. This interruption in generation suggests either scheduled maintenance or temporary system failure.

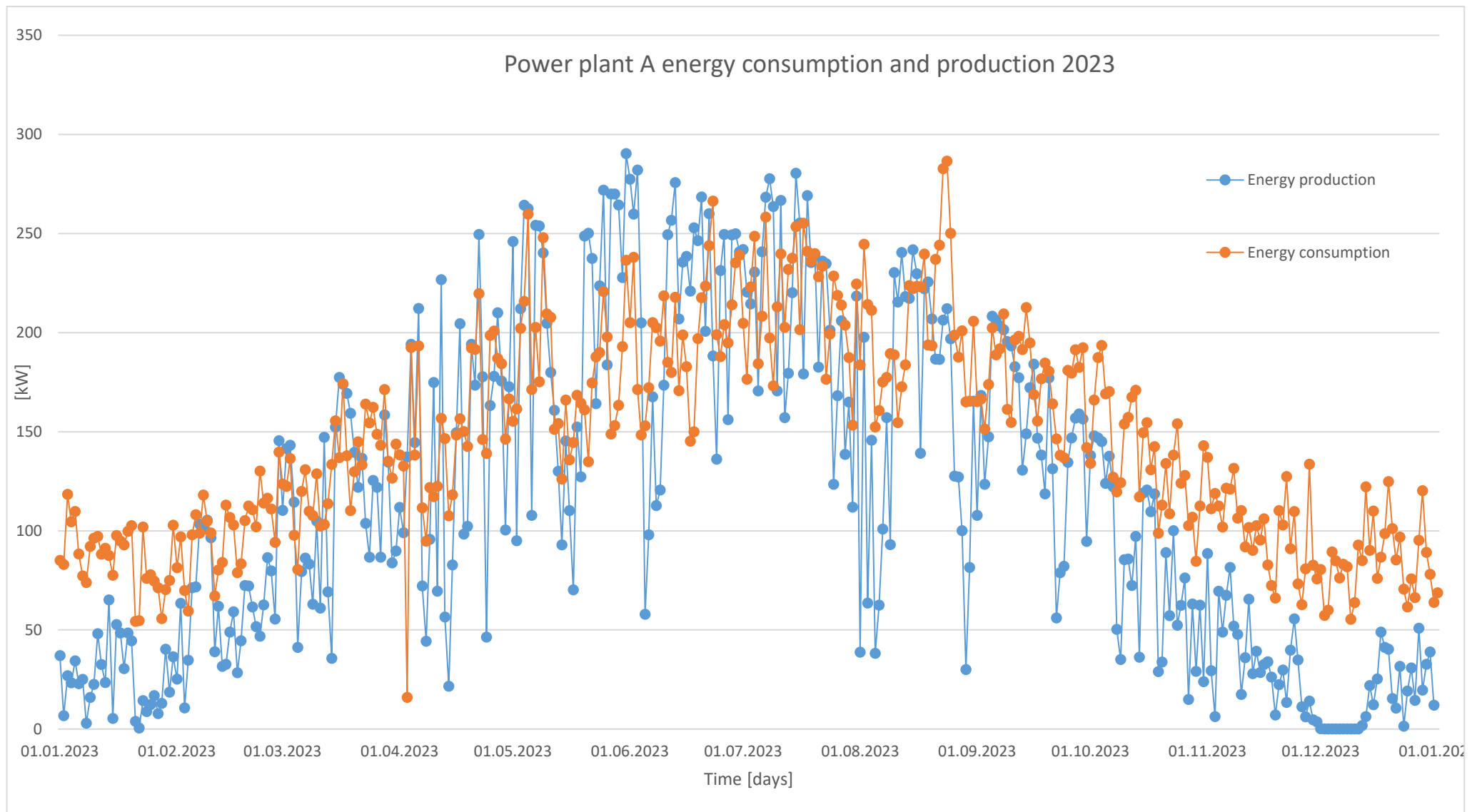


Figure 24: Power plant A energy consumption and production 2023, source: Author

5.2.1.2 The Year 2024

The energy production and consumption trends observed in 2024 closely resemble those of the previous year, following a similar seasonal pattern. As in 2023, January exhibited a production deficit, likely due to limited solar irradiation, shorter daylight hours, and higher energy demand that exceeded the system's generation capacity. This seasonal limitation persisted into February, although the gradual increase in solar intensity led to a progressive improvement in photovoltaic performance.

By mid-March, production levels had risen significantly, with peaks occurring between March 17th and March 20th, reaching a maximum daily output of 182.842 kWh on March 18th. During this period, the photovoltaic system largely covered the facility's energy demand, except for occasional shortfalls attributed to possibly temporary adverse weather conditions. A notable increase in production continued into May, with a peak of 277.38 kWh recorded on May 14th. However, intermittent fluctuations in production, such as a drop to 89.844 kWh on May 25th, suggest the possible impact of varying atmospheric conditions. Unlike production, energy consumption during the spring months remained relatively stable, showing fewer variations. The lower fluctuations in demand may be attributed to the facility's operational stability, as can be seen in Figure 25.

On July 2nd, the generation reached 265.495 kWh, closely matching consumption at 253.447 kWh. This point marked the beginning of a more stable correlation between energy production and demand, which persisted throughout the summer. The power plant was largely able to supply the facility's energy requirements during this period, with occasional shortfalls. However, starting in September, a gradual decline in production became evident while energy consumption remained high or exhibited minor fluctuations. The sustained demand into autumn was likely due to the extended operation of cooling systems driven by persistently warm temperatures. Additionally, the possibility of an increased higher load from additional equipment could explain the temporary rise in consumption, though this trend appeared to be short-lived as demand decreased in late September.

Despite the reduction in solar availability during the autumn months, overall system performance in May and June was improved compared to the previous year, indicating more favourable meteorological conditions. This suggests that while seasonal trends remain a dominant factor, inter-annual variations in weather patterns can influence photovoltaic

efficiency. Additionally, the overall system utilization appears to have been optimized, as the energy surplus periods were more effectively matched with demand compared to 2023.

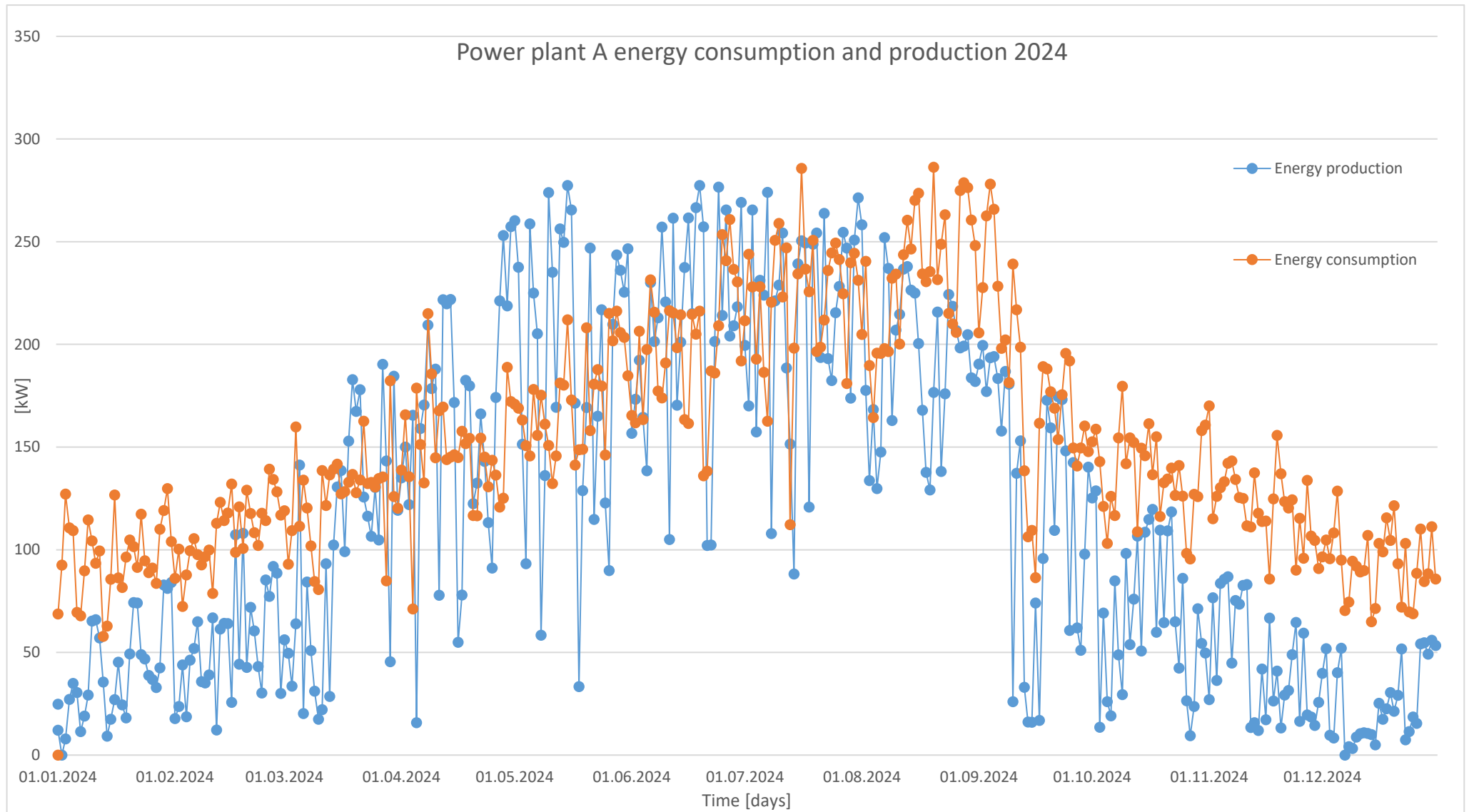


Figure 25: Power plant A energy consumption and production 2024, source: Author

5.2.2 Power Plant Site B

5.2.2.1 The Year 2023

From late March to December 2023, Power Plant B exhibited a distinct pattern of energy consumption and production, reflecting the operational characteristics of the machinery facility it supplies. A key observation is the periodic fluctuation in energy demand, with a noticeable reduction in consumption during weekends. This pattern proves that production activity in the facility follows a standard weekday operational schedule, where CNC machines and other industrial equipment operate at full capacity during workdays and are either partially or completely shut down over the weekends. These fluctuations highlight the cyclical nature of industrial energy demand, which aligns closely with manufacturing processes and work schedules.

In the initial months following commissioning at the end of March, production remained relatively low, likely due to both the system's ramp-up phase and the moderate solar irradiation levels typical of early spring. As the months progressed and solar availability increased, generation levels improved significantly. However, despite this seasonal rise in output, the photovoltaic system was unable to fully meet the facility's energy requirements, as indicated by the persistent gap between production and consumption. The summer months saw the highest production levels, corresponding to peak solar irradiation, yet the facility's substantial and continuous demand meant that grid electricity remained a crucial supplementary energy source.

Another key observation is the presence of frequent and sharp spikes in consumption, often exceeding 1,200,000 kWh on certain weekdays. These peaks suggest periods of heightened industrial activity, likely due to the simultaneous operation of multiple high-demand machines or high-production cycles requiring sustained power input. In contrast, energy production followed a smoother, more predictable seasonal curve, dictated by solar irradiation and weather conditions. The occasional dips in production are likely due to cloud cover, temperature variations, or other meteorological factors.

As autumn approached, a steady decline in production was observed, mirroring the reduction in solar energy availability. While consumption exhibited a slight decrease in the later months of the year, it remained significantly higher than production from the sun, reinforcing the facility's reliance on external electricity sources. The most pronounced

reduction in production occurred in November and December, when daylight hours were the shortest and weather conditions were the least favourable for solar generation. Despite this decline, the cyclical pattern in energy demand persisted, with weekend consumption remaining lower throughout the entire monitoring period in 2023.

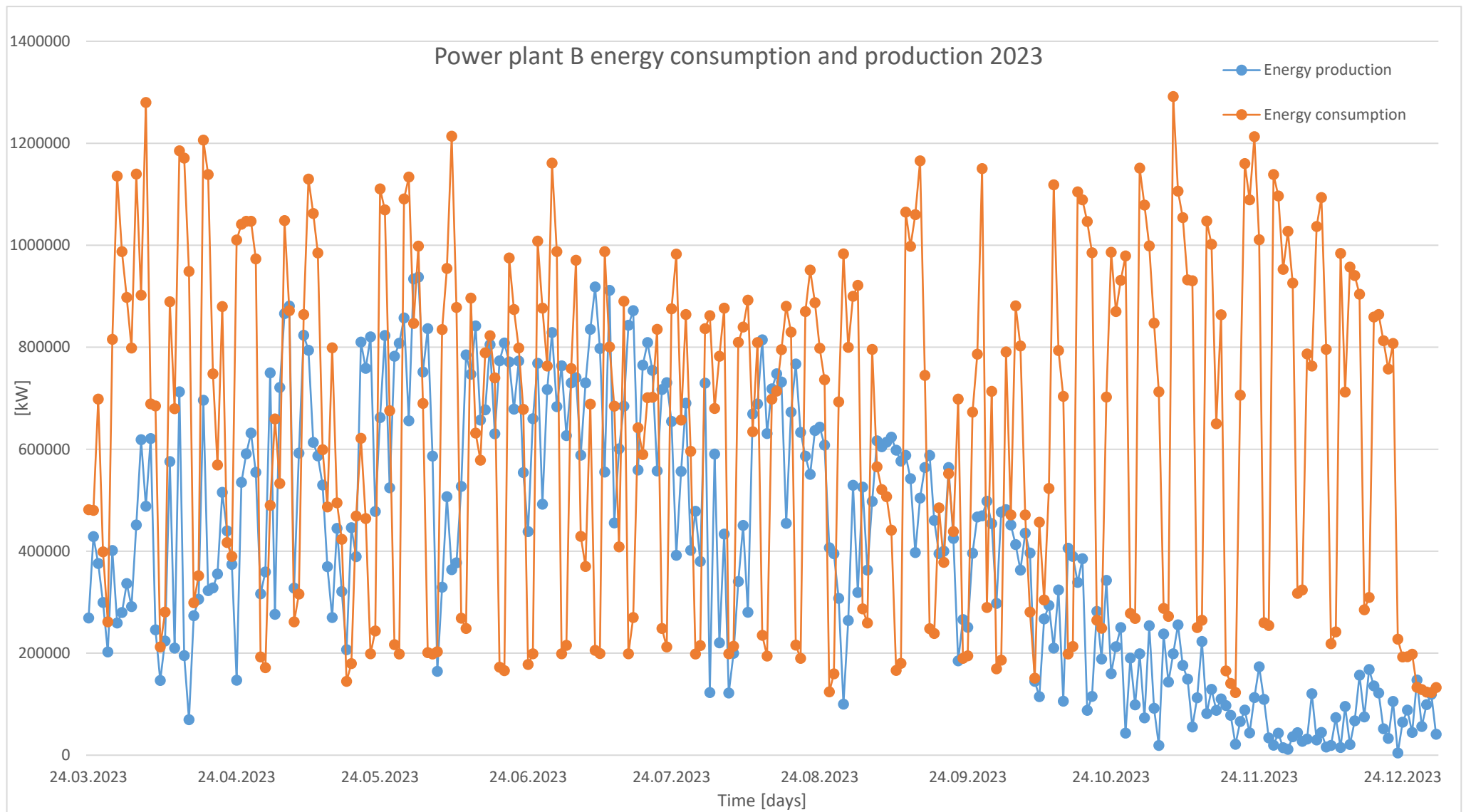


Figure 26: Power plant B energy consumption and production 2023, Source: Author

5.2.2.2 The Year 2024

In 2024, Power Plant B continued to display distinct seasonal variations in energy production, with the characteristic winter decline in output followed by a progressive increase toward peak generation in late spring and summer. Despite improvements in photovoltaic performance, energy consumption remained consistently higher than production throughout the year, emphasizing the facility's continuous and intensive operational demand even in 2024. The consumption profile exhibits the same fluctuations on weekdays as it did in 2023. This pattern reinforces the high and relatively inflexible energy demand of the facility, which remains largely unaffected by seasonal variations in solar availability.

A particularly notable feature of the dataset is the temporary but significant reduction in consumption observed during the summer months, especially in late July and early August. The observed deviation from normal energy usage patterns points to a possible planned reduction in activity, such as a maintenance shutdown of multiple machines, mass employee vacation, or a strategic move to lower operational intensity. This temporary drop in energy demand contrasts sharply with the peak solar generation observed in the same period, exacerbating the mismatch between supply and demand. During this phase, the photovoltaic system not only achieved its highest production values but also frequently generated surplus energy that was not utilized on-site.

As autumn approached, production exhibited the expected decline following the reduction in solar irradiation, though the decrease appeared more gradual compared to 2023. This could indicate a slight shift in seasonal weather conditions. Meanwhile, energy consumption retained its cyclical fluctuations, suggesting that production schedules remained largely unchanged. Even as production waned, consumption remained resilient, demonstrating that the facility's dependence on grid electricity remained a constant factor throughout the year. By the winter months, production levels dropped sharply, further widening the gap between generation and demand. The relatively steady nature of consumption during this period, despite reduced output, underscores the continuous and inflexible energy requirements associated with the facility's operations.

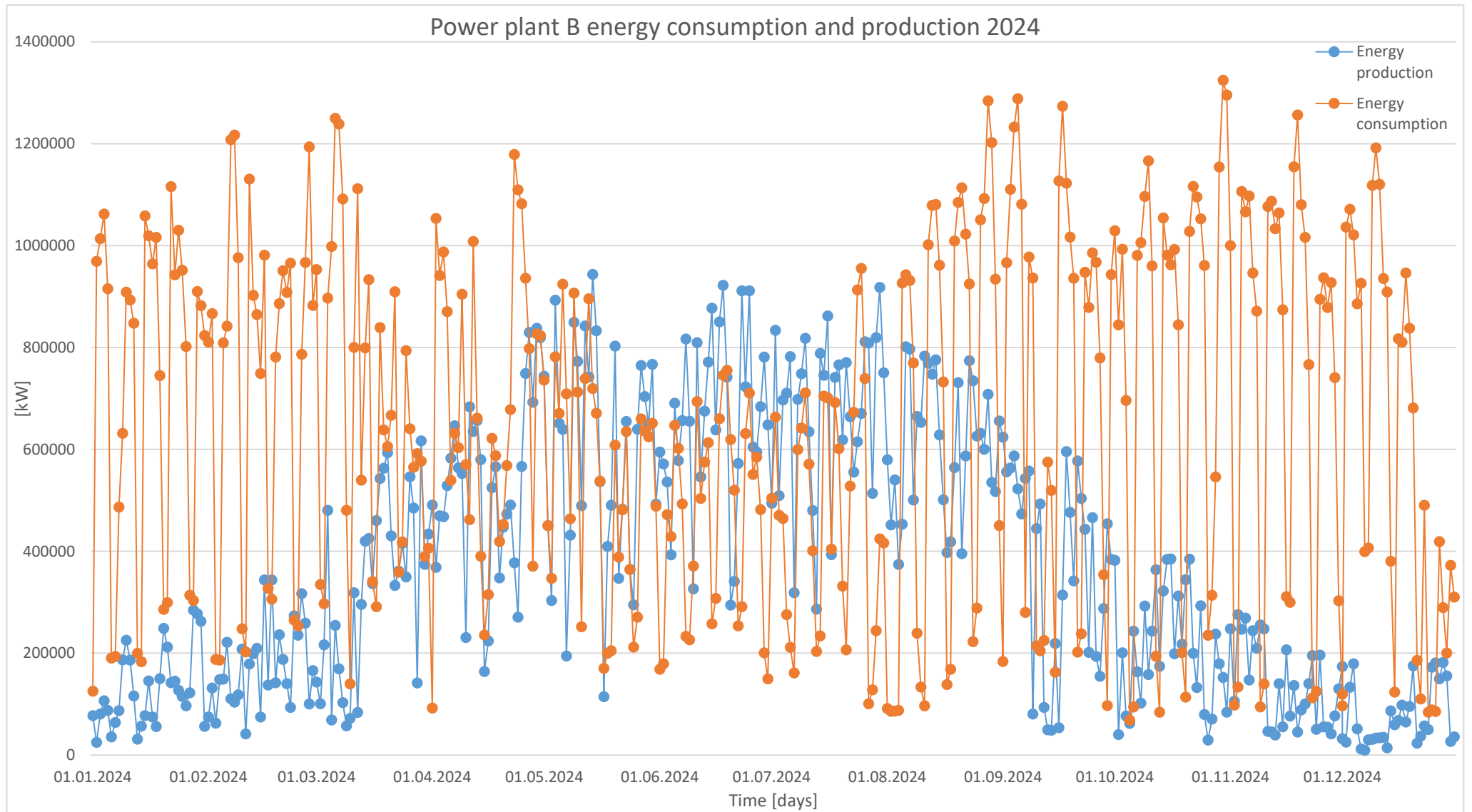


Figure 27: Power plant B energy consumption and production 2024, source: Author

5.3 Technical Calculations and Evaluation of COP and KPI

Table 6: Comparison of Photovoltaic Area Parameters and Solar Irradiation for Power Plants A and B, source: author

	Power plant A	Power plant B
Panel area	2.012 and 2.158 [m ²]	1.955 [m ²]
Total area of panels	195.55 [m ²]	715.53 [m ²]
Panel area per 1 kWp	5.03 and 4.796 [m ² /kWp]	5.13 [m ² /kWp]
Specific Area of PV per 1 kWh produced in 2023	0.00608 [m ² /kWh]	0.00612 [m ² /kWh]
Specific Area of PV per 1 kWh produced in 2024	0.00602 [m ² /kWh]	0.00521 [m ² /kWh]
Specific Surface Area of PV panels per 1 kWh/kWp in 2023	0.242 [m ² /kWh/kWp]	0.854 [m ² /kWh/kWp]
Specific Surface Area of PV panels per 1 kWh/kWp in 2024	0.239 [m ² /kWh/kWp]	0.727 [m ² /kWh/kWp]
Average annual irradiation	1313.4 kWh/m ²	1277.4 [kWh/m ²]
Average daily solar irradiation	3.598 kWh/m ²	3.500 [kWh/m ²]

Power Plant A has a slightly larger individual panel area (2.158 m² per panel for 44 panels and 2.012 m² per panel for 50 panels) compared to Power Plant B (1.955 m² per panel), which is due to the different manufacturers and types of solar panels used. The total installed panel area varies significantly between the two plants: In sum, Power Plant A covers 195.552 m² with 94 panels, while Power Plant B spans 715.53 m² with 366 panels.

The panel area per 1 kWp of installed capacity has been calculated for both plants, with Power Plant A showing a lower area requirement in the second phase (4.796 m²/kWp) compared to the first phase (5.03 m²/kWp). This reduction in panel area required per unit of capacity can be attributed to the higher efficiency of the panels installed during the second phase (20.85%). In comparison, Power Plant B maintains a requirement of 5.13 m²/kWp,

corresponding to its panel efficiency of 21.2%. This difference indicates that the improved efficiency of Power Plant A's second-phase panels directly reduced the area needed to achieve the same output, resulting in a more optimized and space-efficient system. Power Plant B, while having high-efficiency panels, still requires more area per unit of capacity.

In the specific area per 1 kWh produced in 2023, Power Plant A required only 0.00608 m²/kWh produced, while Power Plant B required 0.00612 m²/kWh produced. The small difference between them indicates the previously mentioned nearly relatively close efficiency values of the panels and their conversion of solar energy into usable electricity. In 2024, both plants saw improvements, with Power Plant A reducing its requirement to 0.00602 m²/kWh and Power Plant B decreasing to 0.00521 m²/kWh. The observed improvement is likely attributed to more favorable annual conditions for solar energy generation, including an increased number of sunny days and more stable, optimal temperatures. This trend is evident in both power plants, with Power Plant B demonstrating a more pronounced reduction in specific area per kWh produced.

The specific surface area per kWh/kWp represents the panel surface required to generate one kilowatt-hour per kilowatt of installed capacity. A lower value means the system is more space-efficient, while a higher value indicates that more panel area is required per unit of energy produced. In 2023, Power Plant A exhibited a value of 0.24217 m²/kWh/kWp, whereas Power Plant B required significantly more surface area at 0.854 m²/kWh/kWp. By 2024, both plants demonstrated efficiency improvements, with Power Plant A slightly reducing its value to 0.23940 m²/kWh/kWp, while Power Plant B showed a more substantial enhancement, lowering its value to 0.727 m²/kWh/kWp compared to the year 2023. This could also be associated with previously stated positive conditions.

The average annual solar irradiation for Power Plant A is slightly higher (1313.4 kWh/m²) compared to Power Plant B (1277.4 kWh/m²), representing a 2.8% difference in solar energy input at the location of Power Plant A. Similarly, the average daily solar irradiation is marginally higher for Power Plant A (3.598 kWh/m²/day) than for Power Plant B (3.500 kWh/m²/day). These values were obtained using data from the Global Solar Atlas web application (Solargis, n.d.) and the associated calculation tool. This small difference in irradiation suggests and also indicates that the small difference in geographical location has some impact on the performance of the power plants. However, the tilt angle of the panels plays a slightly more

significant role in energy conversion for these two monitored power plants. As stated, Power Plant A has a tilt angle of 25 degrees, while Power Plant B has a tilt angle of 15 degrees. Additionally, the orientation of the panels relative to the cardinal directions differs, which significantly affects the efficiency of energy production and the overall duration of sunlight exposure. This, in turn, leads to noticeable differences in energy output. The optimal orientation of photovoltaic panels is critical for maximizing energy capture, as panels aligned with the Sun's path throughout the day receive the highest possible amount of solar irradiation. According to Greenvolt Next (2024), the angle of inclination and the azimuthal direction both influence the amount of sunlight captured by the panels, affecting the overall system performance. Panels oriented towards the south in the Northern Hemisphere typically receive the most direct sunlight, leading to higher energy production. Any deviation from this optimal orientation can result in suboptimal energy generation. The precise panel orientations relative to the cardinal directions have already been described in the methodology section.

Table 7: Key Performance Indicators of Power Plants A and B in 2023 and 2024, source: author

	Power Plant A	Power Plant B
Specific Yield for 2023	807.493 [-]	838.043 [-]
Specific Yield for 2024	816.839 [-]	983.976 [-]
Performance ratio 2023	0.615 [-]	0.656 [-]
Performance ratio 2024	0.622 [-]	0.770 [-]
2023 solar panel efficiency under actual operating conditions	12,51 %	12.78 %
2024 solar panel efficiency under actual operating conditions	12,66 %	15.01 %
Weighted Average Panel efficiency	20,10 %	21.2 %
Capacity Utilization Factor 2023	9.22 %	9.57 %
Capacity Utilization Factor 2024	9.32 %	11.23 %

In terms of specific yield, Power Plant A showed a minor increase from 807.493 kWh/kWp in 2023 to 816.839 kWh/kWp in 2024, while Power Plant B displayed a substantial improvement from 838.043 kWh/kWp to 983.976 kWh/kWp. Despite Power Plant B missing almost three months of operation in 2023, its specific yield might have been slightly lower if those months had been included, as January and February typically receive less solar energy due to shorter daylight hours and lower sun angles. The pronounced increase in Power Plant B's yield could suggest that either the plant underwent some kind of system optimization, such as better load management, or external conditions, like the reduction of losses caused by potential shading.

In 2023, Power Plant A had a performance ratio of 0.615. According to IEA-PVPS (2024), a higher PR, closer to 1, indicates minimal losses, while a lower PR suggests greater inefficiencies. Although a performance ratio of 1 represents a theoretical maximum under ideal conditions, well-functioning photovoltaic systems in real-world conditions typically achieve values between 0.75 and 0.85. Between 2023 and 2024, both Power Plants A and B recorded improvements in PR, though the increase was significantly more pronounced for Plant B. In 2024, Power Plant A's PR rose modestly to 0.622, while Power Plant B's performance ratio increased substantially from 0.656 to 0.770. This disparity suggests that Plant B had greater potential for system optimization or operational improvements. In contrast, Plant A may have already been operating closer to its optimal efficiency in 2023, leaving less room for further gains. The marked increase in PR for Plant B is also consistent with its rising specific yield, indicating more effective utilization of its installed capacity. These trends demonstrate that targeted adjustments, likely implemented at Power Plant B, can result in significant efficiency gains, particularly in systems that previously underperformed. In this context, the PR values observed for both plants remain below the expected range for well-optimized systems, suggesting that further improvements may still be achievable.

In 2023, the solar panel efficiency for Power Plant A under actual operating conditions was 12.51%, and it slightly improved to 12.66% in 2024. Power Plant B had a solar panel efficiency of 12.78% in 2023 and a more significant increase to 15.01% in 2024. Year-to-year, if 2024 experienced cooler average temperatures or fewer extreme heat events compared to 2023, the panels in both plants could have operated more efficiently, as photovoltaic system performance is known to decline under high-temperature conditions. (Jordan, 2023) The

efficiency gains observed for Plants A and B under actual conditions likely stem from such temperature and irradiance effects, as well as operational factors. Notably, Plant B shows a larger efficiency improvement than Plant A, which also suggests the previously mentioned differing site conditions or interventions. One reason might be that Plant B experienced a bigger reduction in average module temperature during operation, perhaps due to better ventilation and a milder microclimate.

In 2023, Power Plant A had a Capacity Utilization Factor (CUF) of 9.22%, which slightly increased to 9.32% in 2024. This minimal improvement suggests that Power Plant A was already operating close to its capacity potential, with only minor efficiency gains achieved through system stability and consistent performance. Power Plant B, on the other hand, had a CUF of 9.57% in 2023, but as previously noted, the plant only began operation in March 2023, meaning that the CUF for that year does not account for the typically lower production months of January and February. In 2024, when Power Plant B was operational for the full 12 months, its CUF increased significantly to 11.23%. Part of this increase is due to the inclusion of the missing months.

5.4 Performance Ratio

5.4.1 2023

Both power plants in 2023 exhibited distinct operational differences, particularly in the early months. Power Plant A follows a typical seasonal photovoltaic trend, with its performance ratio (PR) gradually increasing from winter, peaking around June, and declining towards the year's end. This trend is consistent with expected variations in solar irradiation and ambient temperature, which influence photovoltaic efficiency. In contrast, Power Plant B started operations on the 24th of March, as reflected in its initially low PR value, which is a direct result of the commissioning phase.

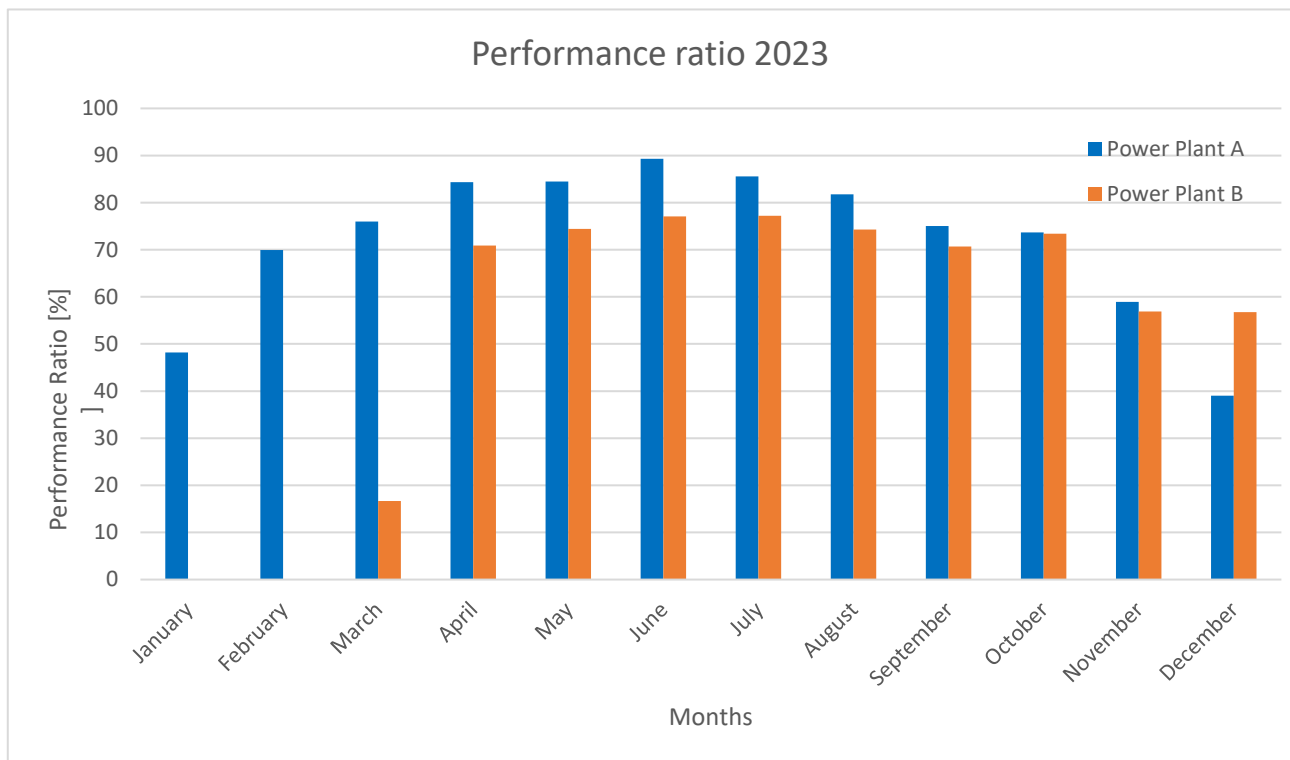


Figure 28: Comparison of Performance Ratio both power plants in 2023, source: Author

From April onwards, Power Plant B demonstrates a significant increase in PR, closely approaching but not fully matching Power Plant A's values. However, despite its significantly larger installed capacity, its PR remains consistently lower than Power Plant A for most of the year. This discrepancy could be attributed to overall non-optimized system settings or localized environmental factors such as the previously mentioned shading. The convergence of PR values in the later months suggests that Power Plant B eventually reached stable operational performance.

The most notable anomaly in Figure 28 is described Power Plant B's near-zero PR in March, which is expected given its late commissioning. However, its ability to achieve stable performance from April onwards suggests an efficient and well-executed commissioning process. Power Plant A, on the other hand, maintains stable and consistent PR throughout the year, reinforcing its well-established operational reliability. While Power Plant B experienced an initial phase of adjustment, its performance stabilized relatively quickly, achieving values comparable to those of Power Plant A by mid-year. This indicates that, aside from the commissioning phase, both plants exhibit similar seasonal PR trends in 2023, reflecting the overarching influence of solar resource availability and system efficiency.

5.4.2 2024

The performance ratio trends for Power Plant A and Power Plant B in 2024 follow expected seasonal variations, with productivity peaking in the summer months and declining in winter due to reduced solar radiation and environmental losses. Compared to 2023, both power plants demonstrate greater stability, particularly Power Plant B, which now has a full dataset for year-round comparison. A key observation is that Power Plant A reaches its peak PR around July, whereas Power Plant B's highest efficiency occurs slightly later, in August and September. This shift, despite both plants being only about a couple of kilometers apart, suggests that localized meteorological differences are unlikely to be the main cause. Instead, system design factors, such as described tilt angles and panel orientations, provide a more plausible explanation. Both of these were introduced in the methodology and materials part of the thesis.

Power Plant A, with a tilt angle of 25° , is better suited for capturing lower sun angles, optimizing performance earlier in the summer, and retaining efficiency in winter months. In contrast, Power Plant B's lower tilt angle of 15° is more optimal for summer conditions, likely explaining its prolonged PR peak into August and September. As noted by Alzahrani and others (2025), the steeper tilt of Power Plant A also contributes to better self-cleaning through rainfall, reducing losses due to soiling and snow buildup in winter. Whereas Power Plant B's flatter angle may lead to a slightly higher accumulation of dirt and snow, causing more significant PR reductions in certain months. This difference is particularly visible in February, where Power Plant B experiences a sharper drop in PR compared to Power Plant A, likely due to panel soiling and snow cover.

A notable anomaly in the 2024 dataset is the pronounced PR decline in October for Power Plant B, diverging from its otherwise strong performance. Given that Power Plant A does not exhibit the same degree of decline, this drop may be attributed to temporary environmental factors or unwanted on-site conditions, which affected Power Plant B's performance. The cause could potentially be seasonal shading from nearby objects or suboptimal system performance under lower autumn irradiation levels, as Power Plant A also experienced a similar, though less pronounced, decline. Such fluctuations warrant further investigation into site-specific influences, including horizon shading analysis and inverter power loading behavior under lower seasonal irradiation.

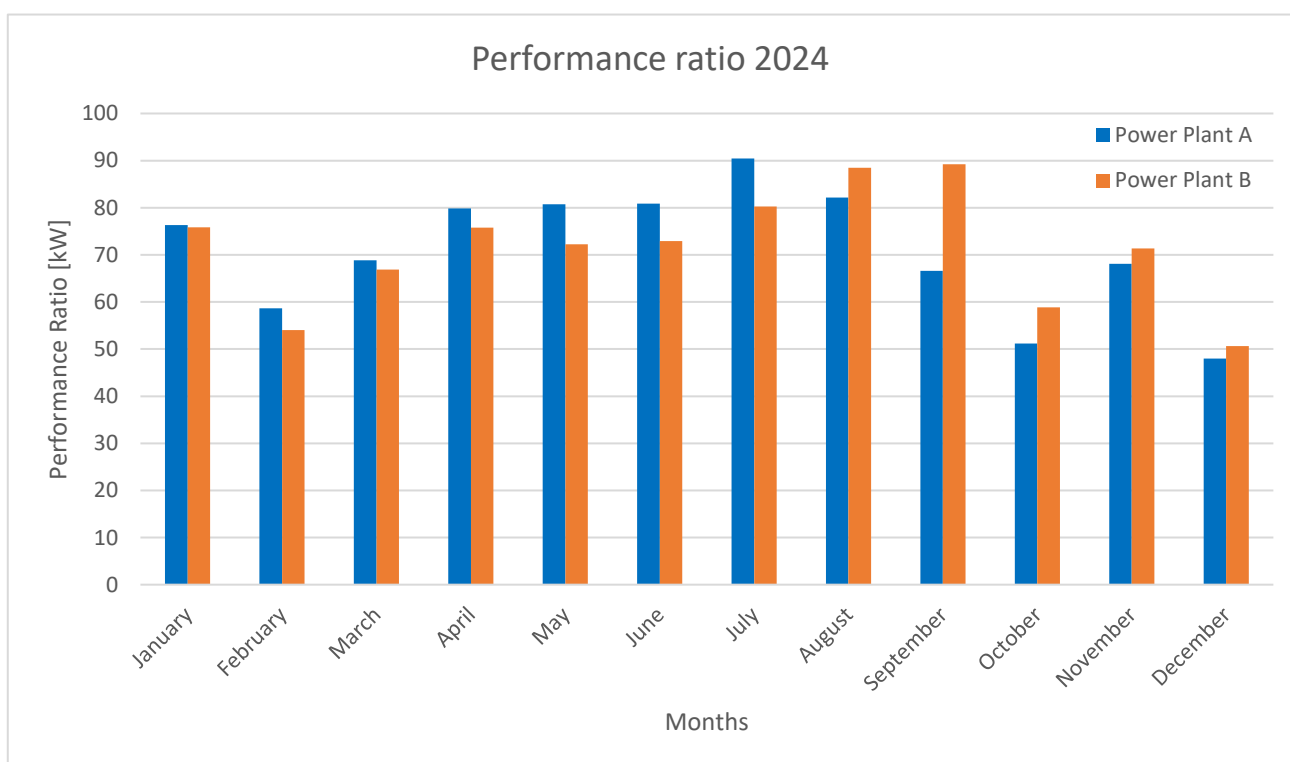


Figure 29: Comparison of Performance Ratio both power plants in 2024, source: Author

However, Power Plant B experiences more pronounced efficiency losses in October and November. Despite this, it reaches slightly higher PR peaks in late summer, likely due to its lower tilt angle, which is better optimized for higher sun elevations. The relatively high PR values recorded by both plants in January may be explained by the effect of colder ambient temperatures on module efficiency, as discussed by Skoplaki and Palyvos (2009), who found that lower temperatures reduce resistive losses and improve energy conversion.

5.5 Comparison Based on PVGIS Simulation Data

The comparison of PVGIS-simulated data with real production values for Power Plant B in 2023 and 2024 reveals strong seasonal correlations but also highlights several month-specific deviations. In both years, the measured data closely follow the simulated trend from May to July, with 2024 production aligning well with PVGIS in both June and July and staying slightly below in May 2023. In contrast, shows slightly lower output than PVGIS across all three months, with a small match-up only in July, confirming that both years approached expected performance during peak irradiation. A major anomaly appears in March 2023, where the actual output is significantly lower than the PVGIS estimate, which is explained by the mentioned fact that the plant only started operation on March 24th, leaving most of the month without production.

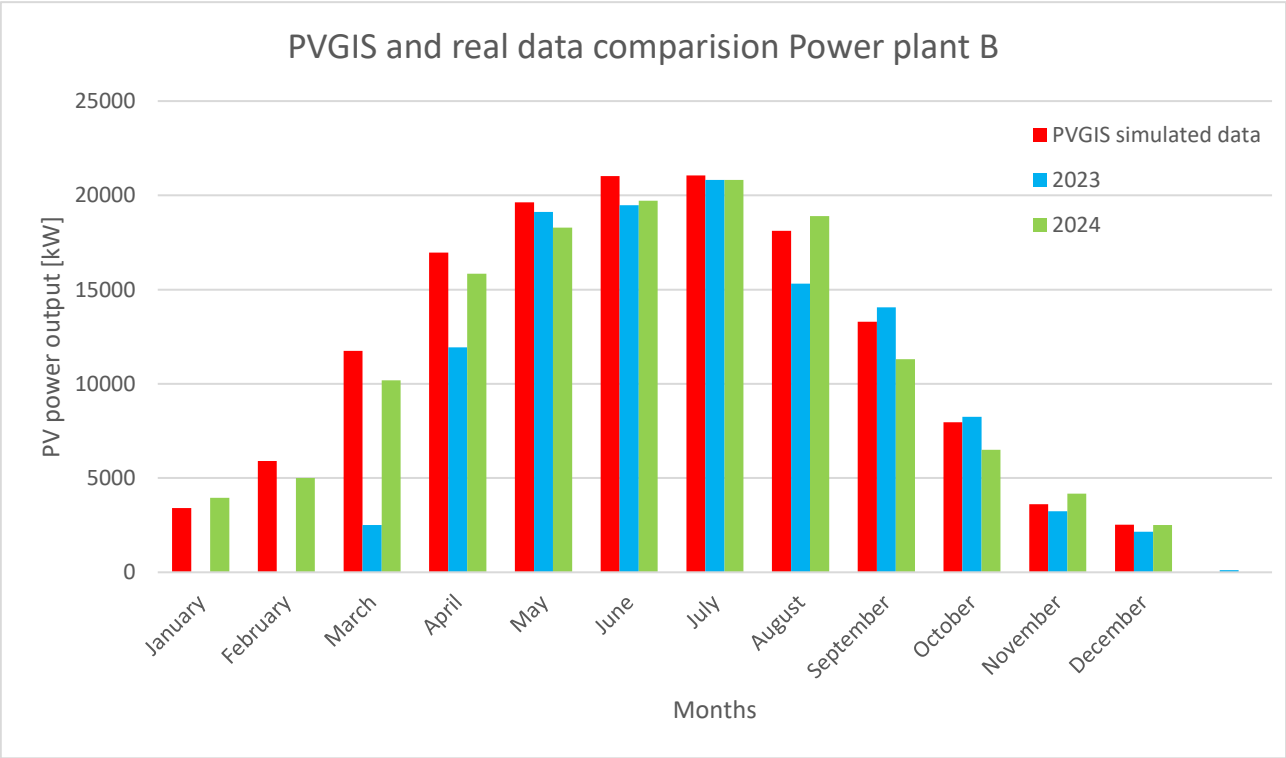


Figure 30: PVGIS and real data comparison of Power plant B, source: Author

March 2024 falls slightly below the simulation, likely reflecting early spring conditions. April 2023 remains well below the PVGIS estimate, while 2024 stays modestly under it. August 2023 shows a clear deficit, while 2024 slightly exceeds the model, pointing to interannual climatic variability. September 2023 notably exceeds PVGIS, suggesting exceptionally favorable conditions that month, while 2024 falls just below the estimate. In October, both years show underperformance, with 2024 deviating more significantly by approximately 2,000

kWh, indicating seasonal shading or environmental losses not fully accounted for in the model simulation. In the winter months, notably November and December 2023 remains slightly below PVGIS, while 2024 shows performance that closely matches or slightly exceeds the simulation, highlighting variability in winter irradiation and site-specific conditions. According to a study by Walker and Balderrama (2021), such deviations may result from factors like snow accumulation, prolonged inverter wake-up times during low-light conditions, or persistent overcast skies, all of which could contribute to the observed seasonal underperformance.

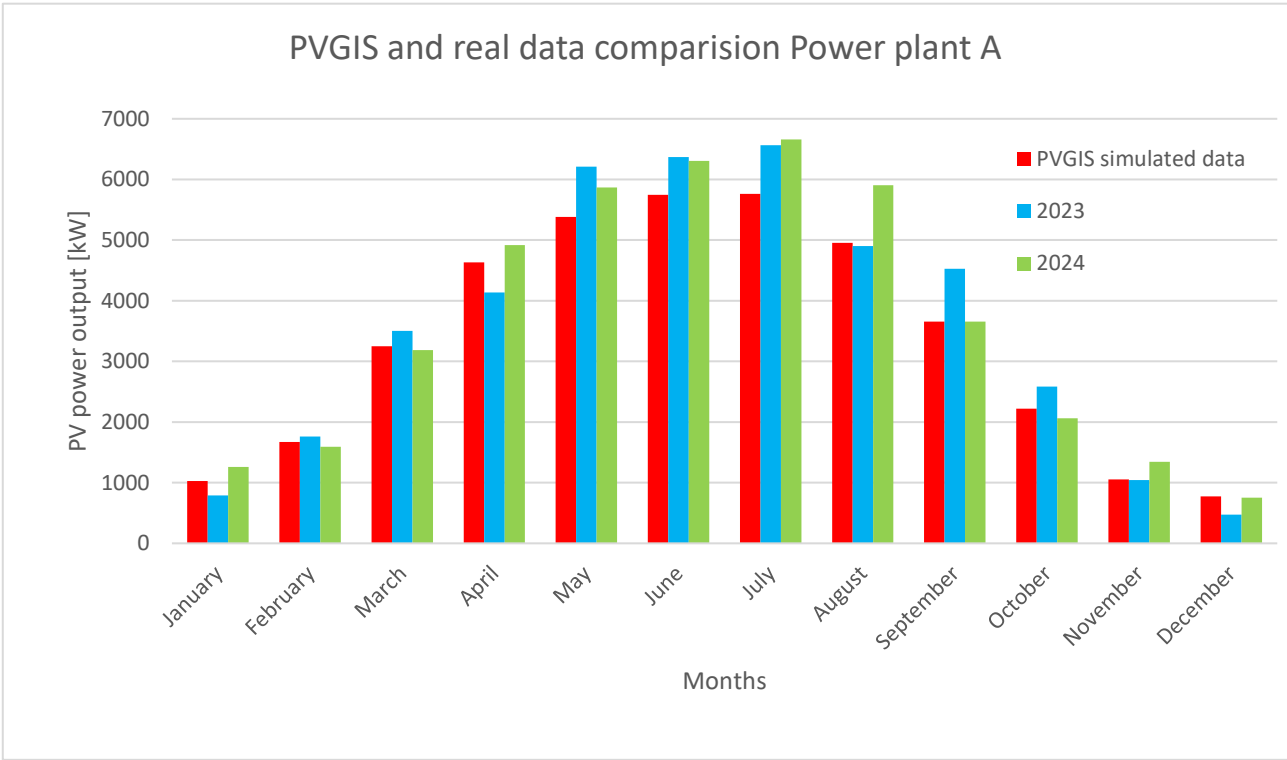


Figure 31: PVGIS and real data comparison of Power plant A, source: Author

The comparison between PVGIS-simulated photovoltaic output and real production data from Power Plant A in 2023 and 2024 also reveals clear seasonal patterns along with moderate deviations. During the winter months (January and February), actual values in both years are slightly above or close to the simulated estimates. In March and April, the alignment remains, with only fluctuations, especially in the year 2023. From May to July, both years consistently exceeded PVGIS projections, particularly in May and July, where the real output was notably higher. This indicates that the power plant outperformed the long-term average conditions represented by the model, likely due to locally favorable weather or optimal operational conditions during this period. In August 2023, production aligns closely with the simulation, while 2024 shows higher values, highlighting interannual climatic variability.

September marks a shift, with 2024 closely matching the PVGIS estimate, while 2023 exceeds it. This divergence continues in October as 2023 exceeds the model output, whereas 2024 remains slightly below, reflecting sensitivity to short-term weather conditions. Finally, in November and December, measured outputs in both years are close to the PVGIS values, with only small differences. Overall, the PVGIS model provides a reliable baseline, with real data generally tracking its seasonal trend.

5.6 Operational Optimization

Based on the monitored performance of Power Plant A, the system demonstrates generally stable operation with consistently high utilization efficiency under standard conditions, however, surplus production during spring months indicates room for further optimization in energy management and load shifting. In 2023, the performance ratio reached 0.615 and improved slightly to 0.622 in 2024, confirming stable long-term performance. The self-consumption rate remained high, with 74.84% of produced energy used on-site and a relatively low grid dependency of 41.35%. These values reflect a well-balanced system aided by a 25° tilt angle and south-facing orientation of one-half of the installed panels, which align effectively with the solar resource availability and the facility's moderate energy demand. Despite the generally optimized operation, performance data reveal a recurrent seasonal surplus during spring, particularly in April and May, when production often exceeds consumption. During these periods, excess energy amounting to over 25% of the total generation is exported to the grid. This indicates the potential for adaptive load management strategies.

In facilities where cooling systems operate continuously to maintain stable temperatures, such as refrigerated storage or cooled process tanks, minor adjustments in control logic can shift energy-intensive operations (e.g., compressor cycling, thermal buffering) to align with periods of peak solar generation. By modulating the timing of these functions within acceptable thermal tolerances, the system could increase on-site utilization without affecting operational stability. Such strategies have been shown to improve PV integration in facilities with steady thermal loads. (Abdmouleh, Alammari, & Gastli, 2015). The plant's tilt and exposure allow for sufficient self-cleaning, and the system shows no evidence of production collapse. Therefore, major upgrades like storage integration or panel

replacement are not currently justified. Improvements should instead focus on software-level optimization

In contrast, Power Plant B faces more complex operational conditions. As described, it serves a high-load industrial facility with relatively inflexible demand patterns and is equipped with a flatter 15° panel tilt. In 2023, the PR was 0.656, increasing significantly to 0.770 in 2024. Similarly, the Capacity Utilization Factor (CUF) rose from 9.57% to 11.23%. These improvements suggest that some optimizations were done between monitored years. Nonetheless, inefficiencies remain. A persistent issue is weekend overproduction, as industrial activity drops while solar generation remains high. In both years, weekends exhibited significant surplus generation. The research conducted by Luthander, Widén, Nilsson, & Palm (2015) underscores the advantages of integrating medium-scale battery energy storage systems (BESS) in such contexts. A battery storage system capable of storing 10–15% of daily generation, based on observed recurring overproduction on weekends and during high-insolation periods. Although no detailed simulation was conducted, this value corresponds with typical oversupply trends observed in summer months and could reduce grid export by an estimated 5–10%. It could capture this excess and discharge it during weekday peak demand, reducing peak grid imports and improving the utilization of solar energy. In October 2024, a sharp drop in production and PR was recorded, while Power Plant A, located nearby, did not exhibit such a decline. This could prove local site-specific issues, such as heavy soiling. The installation of basic soiling sensors would enable early detection of excessive dirt accumulation, allowing timely cleaning and reducing energy yield losses due to oversoiling. Scheduled cleaning during September or early October can further help mitigate recurring autumn-related losses.

To reduce mismatch losses during summer periods of reduced operational activity, the facility could implement several targeted measures. These include rescheduling partial operations to coincide with peak photovoltaic generation days and temporarily activating non-essential systems, such as compressed air units, HVAC balancing, or auxiliary processes with flexible demand profiles. As highlighted by Luthander, Widén, Nilsson, & Palm (2015), such load-shifting strategies are among the most effective methods for increasing on-site photovoltaic utilization in facilities with inflexible or partially flexible demand. Installing a load-shifting controller capable of dynamically responding to real-time solar output would further

enhance system responsiveness. Even partial alignment of internal loads with available generation can significantly reduce midday energy export, especially during high-irradiance periods typical of late July and early August. These interventions require minimal infrastructure changes and can be integrated into existing energy management systems. Additionally, automated scheduling based on irradiance forecasts could optimize demand timing and further improve self-consumption efficiency.

Winter performance also warrants attention. The flat panel tilt reduces the effectiveness of self-cleaning mechanisms under snow and fog conditions, which exacerbates seasonal energy losses. According to Law, Jones & Walls (2023), simple mechanical interventions, such as hydrophobic surface coatings or moderate tilt adjustments, have shown a potential to improve winter energy yield, particularly in installations exposed to snow accumulation and soiling. These solutions are relatively easy to implement and can enhance seasonal efficiency without requiring major structural changes.

In summary, Power Plant A requires only targeted software-based refinements to maintain optimal operation. Power Plant B, however, would benefit from a combination of operational planning and moderate hardware-based upgrades, including storage, soiling mitigation, and environmental performance monitoring to address persistent mismatches and seasonal inefficiencies.

6 Conclusion

This diploma thesis focused on the analysis and evaluation of crucial operational parameters of two photovoltaic power plants, designated as Power Plant A and Power Plant B. The aim was to assess their real-world performance using crucial operating parameters and other technical parameters over a two-year monitoring period (2023 and 2024). The theoretical part covered the origins and historical development of photovoltaics, along with the fundamental principles of PV systems and the role of key components such as panels, inverters, and batteries. The practical part provided a detailed comparison of both systems, considering their design configuration, operational environment, and energy demand characteristics.

Power Plant A, characterized by a smaller installed capacity (39.8 kWp), higher panel tilt (25°), and relatively stable consumption driven by a food-processing facility, exhibited consistently high self-consumption rates (74.84%), low grid dependency (41.35%), and a stable performance ratio (0.615 to 0.622). Its system design proved well-aligned with its operational demand, with no evidence of some major performance losses or hardware failure. However, regular surplus energy generation in spring indicates room for further optimization. Without requiring additional infrastructure, this mismatch can be addressed through adaptive load-shifting strategies tailored to central cooling systems by timing compressor cycles and thermal buffering to coincide with midday solar peaks, thereby increasing real-time consumption and reducing grid exports.

In contrast, Power Plant B operates at a significantly larger scale (139.44 kWp), serving a high-demand industrial site with fluctuating energy loads and a flatter 15° panel tilt. While improvements were observed across all monitored indicators, most notably the rise in performance ratio from 0.656 to 0.770 and the increase in capacity utilization factor from 9.57% to 11.23%, several seasonal inefficiencies remain. Based on the analysis presented in this thesis, persistent weekend overproduction, resulting from reduced industrial activity, was identified as a recurring pattern. To address this issue, it is recommended to implement partial energy storage or intelligent load management. Specifically, the installation of a medium-sized battery energy storage system (BESS) capable of storing approximately 10–15% of the daily energy output, or the use of a dynamic load-shifting controller, could significantly reduce unnecessary grid exports and improve overall system efficiency during periods of low demand.

Based on the findings of this thesis, additional inefficiencies were identified during autumn 2024, most notably in October, when Power Plant B experienced a significant drop in production not observed at Power Plant A. This suggested the influence of localized environmental factors, such as soiling or partial shading. To address these losses, it is recommended to install basic soiling sensors and introduce a scheduled cleaning cycle before the onset of autumn. To reduce winter losses, the use of hydrophobic coatings was recommended as a simple, non-invasive improvement.

Overall, the findings demonstrate that both systems are operating almost efficiently within their designed constraints. Power Plant A is already close to optimal, requiring only minor software-level refinements. Power Plant B, while demonstrating significant improvement between monitored years, presents additional opportunities for performance enhancement through a combination of mechanical adjustments, environmental monitoring, and operational planning. These recommendations are practical, scalable, and aligned with real-world constraints, offering valuable pathways for further increasing the energy independence of industrial facilities.

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